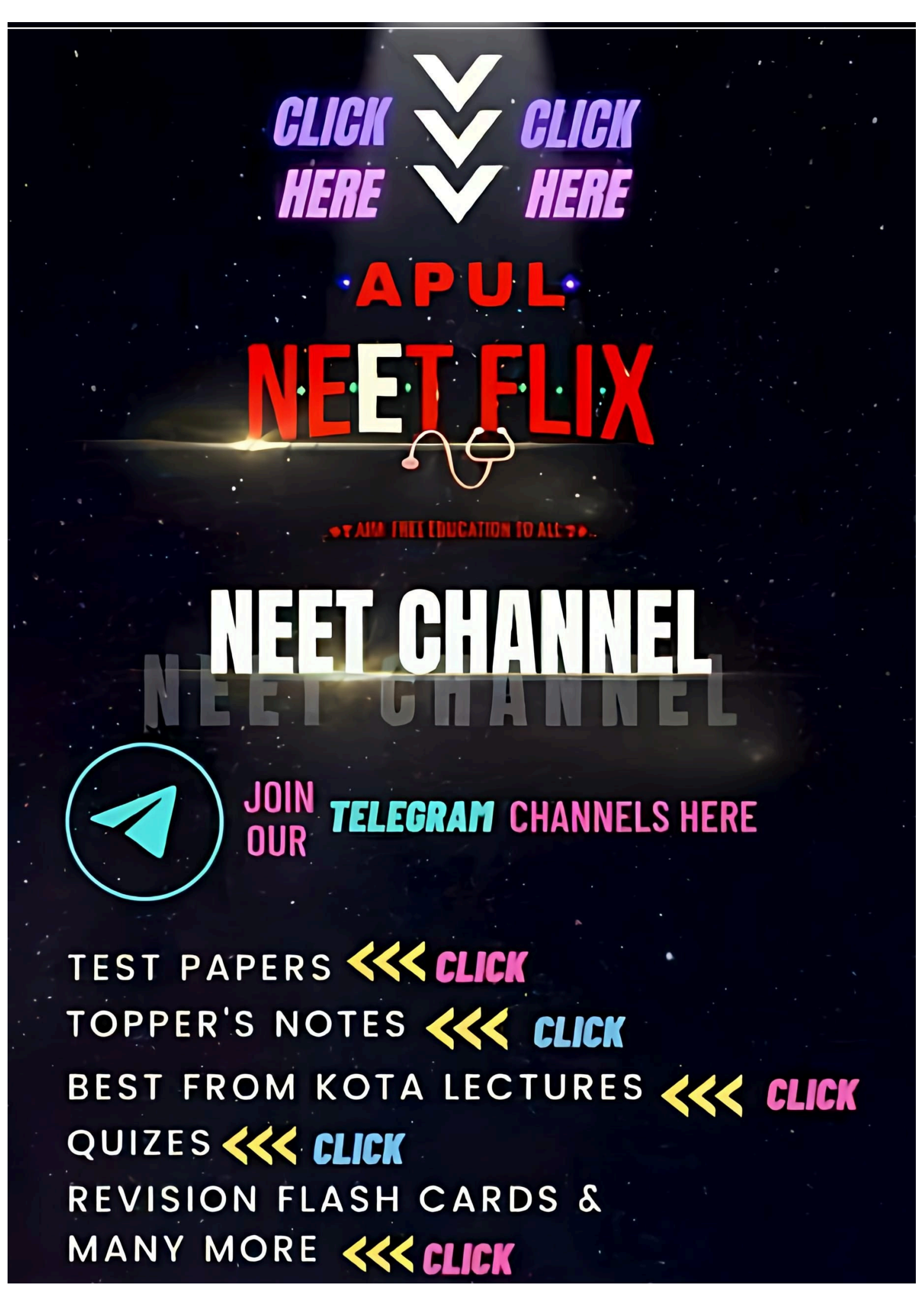
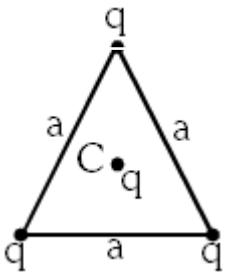

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PHYSICS

1) Four equal charges of magnitude q are placed as shown in figure. Now the charge at center C is taken to infinite slowly, then work done by external force will be -



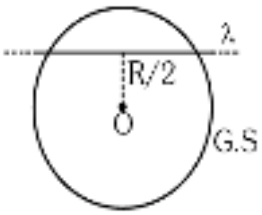
(1) $\frac{-3kq^2}{a}$

(2) $\frac{-3\sqrt{3}kq^2}{a}$

(3) $\frac{-3kq^2}{a}(\sqrt{3}+1)$

(4) $\frac{3kq^2}{a}(\sqrt{3}+1)$

2) Find flux from G.S. (Radius of sphere is R)



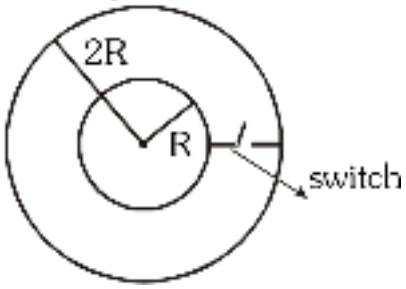
(1) $\frac{\sqrt{3}\lambda R}{2\epsilon_0}$

(2) $\frac{\sqrt{3}\lambda R}{\epsilon_0}$

(3) $\frac{\lambda R}{\epsilon_0}$

(4) $\frac{2\lambda R}{\epsilon_0}$

3) Net positive charge on inner spherical conductor is 3Q and on outer spherical conductor is Q.



When switch is closed :-

(1) Charge will flow from inner shell to outer shell

(2) Charge will flow from outer shell to inner shell

(3) Charge will not flow

(4) None of these

4) Deutron and α-particle is at 1Å distance in air. Calculate the magnitude of electric field intensity at α-particle due to deutron :-

(1) Zero

(2) 2.88×10^{11} N/C

(3) 1.44×10^{11} N/C

(4) 5.76×10^{11} N/C

5) One picocoulomb of charge is distributed on a ring of diameter 1 mm. The potential at the centre of the ring is:-

(1) zero

(2) 18 volt

(3) 9 volt

(4) –14.4 volt

6) Two protons release from mutual distance r, find the maximum velocity they can achive (assume charge of proton = e) :-

(1) $v = e\sqrt{\frac{K}{m_p r}}$

(2) $v = e\sqrt{\frac{2K}{m_p r}}$

(3) $v = e\sqrt{\frac{K}{2m_p r}}$

(4) None of these

7) Three charges 4q, – 2q, and 2q are placed at distance r in xy plane from origin such that the lines joining the charges to origin are making angles 30°, 150° and 330° respectively from positive x axis then electric field at origin is :-

(1) $\frac{kq}{r^2}$

(2) $\frac{4kq}{r^2}$

(3) $\frac{4\sqrt{3}kq}{r^2}$

(4) $\frac{2\sqrt{3}kq}{r^2}$

8) In the given figure, a point charge q is placed at the center of circular surface C and flux through

surface A is ϕ then flux through curved surface B is :



(1) $\frac{q}{\epsilon_0} - \phi$

(2) $\frac{q}{2\epsilon_0} - \phi$

(3) $\frac{1}{2}\left(\frac{q}{\epsilon_0} - \phi\right)$

(4) Zero

9) A soap bubble is charged to a potential of 16 V. Its radius is then doubled. The potential of the bubble will now be :

(1) 2 V

(2) 4 V

(3) 8 V

(4) 16 V

10) A ring of radius R carries a charge +q. A test charge $-q_0$ is released on its axis at a distance $\sqrt{3}R$ from its centre. How much kinetic energy will be acquired by the test charge when it reaches the centre of the ring ?

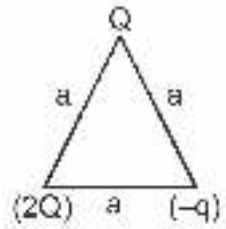
(1) $\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{R}$

(2) $\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{2R}$

(3) $\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{\sqrt{3}R}$

(4) $\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{3R}$

11) If electric potential energy of given system is positive then :



(1) $2Q > 3q$

(2) $2Q < 3q$

(3) $Q > 2q$

(4) $Q > 3q$

12) Two identical metal A and B spheres having charges 10 μC and 2 μC placed in contact then what amount of mass will be transferred :-

(1) 4×10^{-6} kg

(2) 8.2×10^{-8} kg

(3) 6.5×10^{-10} kg

(4) 2.2×10^{-17} kg

13) The current in the windings of a toroid is 2.0A. There are 400 turns and the mean circumferential length is 40cm. If the inside magnetic field is 1.0T, the relative permeability is near to :-

(1) 100

(2) 200

(3) 300

(4) 400

14) An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force of electron is zero

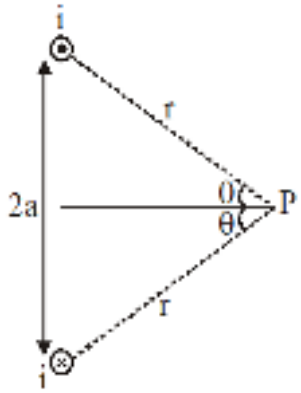
(1) positive z- axis

(2) negative z-axis

(3) positive y-axis

(4) negative y-axis

15) Two long parallel wires situated at a distance 2a are carrying equal current 'i' in opposite direction as shown in figure. The value of magnetic field at a point P situated at equal distances from both the wires will be :



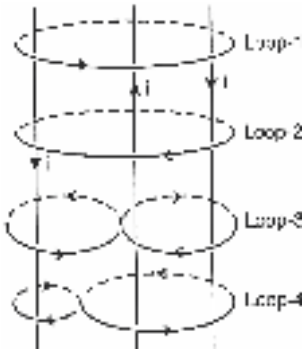
(1) $\frac{\mu_0 ia}{\pi r}$

(2) $\frac{\mu_0 ia^2}{\pi r}$

(3) $\frac{\mu_0 ia^2}{\pi r^2}$

(4) $\frac{\mu_0 ia}{\pi r^2}$

16) Three wires are carrying same current i in given direction. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in the figure :



	List-I		List-II
(P)	Along closed Loop-1	(1)	$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
(Q)	Along closed Loop-2	(2)	$\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
(R)	Along closed Loop-3	(3)	$\oint \vec{B} \cdot d\vec{\ell} = 0$
(S)	Along closed Loop-4	(4)	$\oint \vec{B} \cdot d\vec{\ell} = 3\mu_0 i$

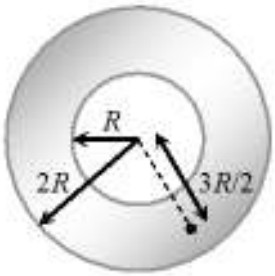
(1) P-2 → Q-1 → R-2 → S-1

(2) P-1 → Q-2 → R-1 → S-2

(3) P-2 → Q-1 → R-3 → S-4

(4) P-4 → Q-4 → R-3 → S-3

17) Figure shows the cross-sectional view of the hollow cylindrical conductor with inner radius 'R' and outer radius '2R', cylinder carrying uniformly distributed current i along it's axis. The magnetic



induction at point 'P' at a distance $\frac{3R}{2}$ from the axis of the cylinder will be :-

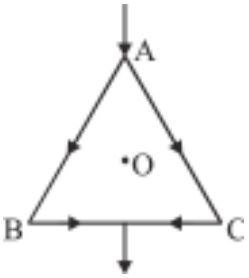
(1) Zero

(2) $\frac{5\mu_0 i}{72\pi R}$

(3) $\frac{7\mu_0 i}{18\pi R}$

(4) $\frac{5\mu_0 i}{36\pi R}$

18) An equilateral triangle is made by uniform wires AB, BC, CA. A current I enters at A and leaves from the mid-point of BC. If the length of each side of the triangle is L, the magnetic field B at the



centroid O of the triangle is :-

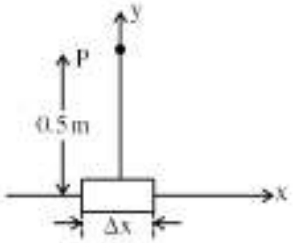
(1) $\frac{\mu_0}{4\pi}\left(\frac{4I}{L}\right)$

(2) $\frac{\mu_0}{2\pi}\left(\frac{4I}{L}\right)$

(3) $\frac{\mu_0}{4\pi}\left(\frac{2I}{L}\right)$

(4) Zero

19) An element $\Delta l = \Delta x\hat{i}$ is placed at the origin and carries a large current I = 10A. The magnetic field on the y-axis at a distance of 0.5 m from the elements Δx of 1 cm length is :



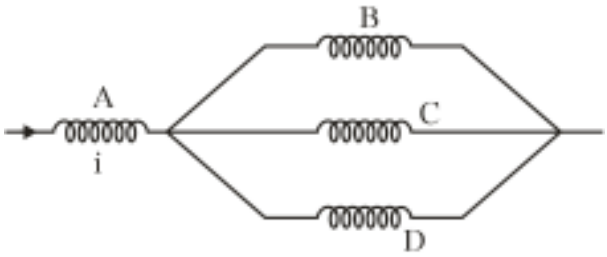
(1) $4 \times 10^{-8}T$

(2) $8 \times 10^{-8}T$

(3) $12 \times 10^{-8}T$

(4) $10 \times 10^{-8}T$

20) Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is 3T, the field at the center of C would be : (Assume that the magnetic field is confined with in the volume of respective solenoid).



(1) 12T

(2) 6T

(3) 9T

(4) 1T

21) The radius of a circular coil is R and it carries a current of I ampere. The intensity of magnetic field at a distance x on the axis from the centre (x >>R) will be :-

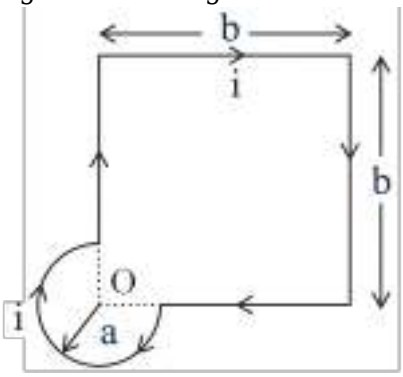
(1) $B = \frac{\mu_0}{2}\frac{IR^2}{x^2}$

(2) $B = \frac{\mu_0 IR^2}{2x^3}$

(3) $B = \frac{\mu_0 IR}{2x^2}$

(4) $B = \frac{\mu_0 IR}{2x^3}$

22) Find the magnitude of magnetic field at O (centre of the circular part) of the current carrying coil as shown.



(1) $\frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{a}+\frac{\sqrt{2}}{b}\right)$

(2) $\frac{\mu_0 i}{2\pi}\left(\frac{3\pi}{a}+\frac{\sqrt{2}}{b}\right)$

(3) $\frac{\mu_0 i}{2\pi}\left(\frac{\pi}{3a}+\frac{3}{\sqrt{2}b}\right)$

(4) $\frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right)$

23) The magnetic field at the centre of an equilateral triangular loop of side 2L and carrying a current i is :-

(1) $\frac{9\mu_0 i}{4\pi L}$

(2) $\frac{3\sqrt{3}\mu_0 i}{4\pi L}$

(3) $\frac{2\sqrt{3}\mu_0 i}{\pi L}$

(4) $\frac{3\mu_0 i}{4\pi L}$

24) An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction tripled, the radius will become

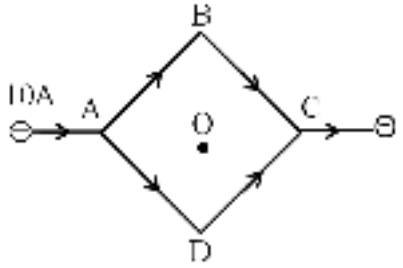
(1) $\frac{3R}{2}$

(2) $\sqrt{\frac{3}{2}}R$

(3) $\sqrt{\frac{2}{9}}R$

(4) $\sqrt{\frac{4}{3}}R$

25) Figure shows a square loop with edge length 'a' meter. The resistance of wire ABC is 'r' Ω and



that of ADC is '2r' Ω. Magnetic field at the centre of the loop is :-

(1) $\frac{10\sqrt{2}\mu_0}{3\pi a}\otimes$

(2) $\frac{10\sqrt{2}\mu_0}{3\pi a}\odot$

(3) $\frac{\sqrt{2}\mu_0}{6\pi a}\otimes$

(4) $\frac{\sqrt{2}\mu_0}{6\pi a}\odot$

26) If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be :-

(1) only inside the pipe

(2) only outside the pipe

(3) both inside and outside the pipe

(4) neither inside nor outside the pipe

27) The magnetic field on the axis of a circular loop of radius 100 cm carrying current $I = \sqrt{2}\,A$, at point 1 m away from the centre of the loop is given by:

(1) 3.14×10^{-7} T

(2) 6.28×10^{-7} T

(3) 3.14×10^{-4} T

(4) 6.28×10^{-4} T

28) Ionized hydrogen atoms and α-particles with same momenta enters perpendicular to a constant magnetic field B. The ratio of their radii of their paths $r_H : r_\alpha$ will be

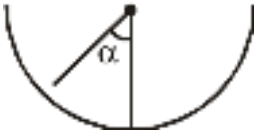
(1) 2 :1

(2) 1 : 2

(3) 4 : 1

(4) 1 : 4

29) A insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the surface and the insect is $\frac{1}{3}$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given :–



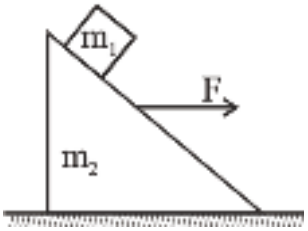
(1) cot α = 3

(2) tan α = 3

(3) sec α = 3

(4) cosec α = 3

30) A block of mass m_1 lies on a wedge on mass m_2. Friction is absent everywhere. An external horizontal constant force F is applied on the wedge as shown in the figure. The value of F for which no relative slipping takes place between blocks is:



(1) m_1 g tan θ

(2) m_2 g tan θ

(3) $[m_1 + m_2]$ g tan θ

(4) $[m_1 - m_2]$ g tan θ

31) A body takes time t to reach the bottom of an inclined plane of angle θ with the horizontal if plane is smooth. If the plane is made rough, time taken now is 2t. The coefficient of friction of the rough surface is :-

(1) $\frac{3}{4}$ tan θ

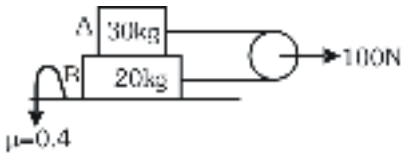
(2) $\frac{2}{3}$ tan θ

(3) $\frac{1}{4}$ tan θ

(4) $\frac{1}{2}$ tan θ

32)

If there is no friction between blocks, then acceleration of block A will be :



(1) 0

(2) $\frac{5}{3}$m/s^2

(3) $\frac{5}{6}$m/s^2

(4) 2 m/s^2

33) A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 5 m/s^2. What would be the reading on the scale (g = 10 m/s^2) :-

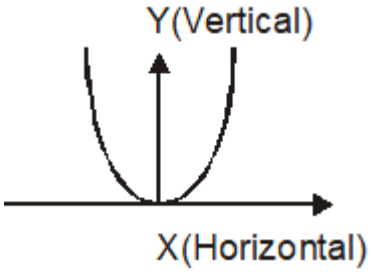
(1) 400 N

(2) 800 N

(3) 1200 N

(4) Zero

34) A parabolic bowl with its bottom at origin has the shape y = x^2/20. Here x and y are in meter. The maximum height at which a small mass m can be placed on the bowl without slipping (coefficient of static friction is 0.5) is



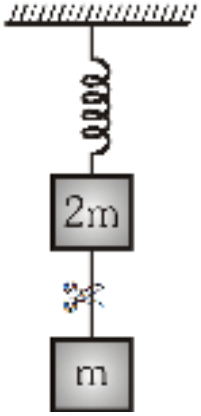
(1) 2.5 m

(2) 1.25 m

(3) 1.0 m

(4) 4.0 m

35) System shown in figure is in equilibrium and at rest. The spring and string are massless, now the string is cut. The acceleration of mass 2m and m just after the string is cut will be :–



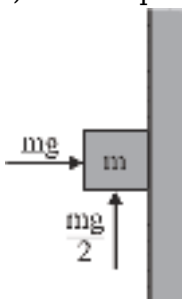
(1) $\frac{g}{2}$ upwards, g downwards

(2) g upwards, $\frac{g}{2}$ downwards

(3) g upwards, 2g downwards

(4) 2g upwards, g downwards

36) A block pressed against the vertical wall is in equilibrium. The minimum coefficient of friction is



:-

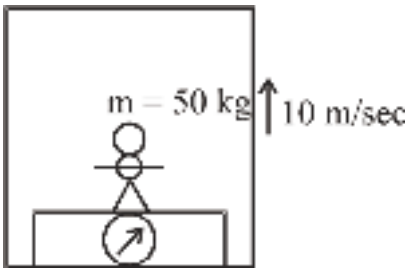
(1) 0.4

(2) 0.2

(3) 0.5

(4) None of these

37) The reading of weighing machine in kg-wt :-



(1) 50

(2) 100

(3) 500

(4) 150

38) A uniform chain of length 2L is hanging in equilibrium position. If end B is given a slightly downward displacement, the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible. The acceleration of end B when it has been displaced by distance x, is :-



(1) $\frac{x}{L}g$

(2) $\frac{2x}{L}g$

(3) $\frac{x}{2g}$

(4) g

39) Two persons A and B are sitting in two cars 1 and 2 such that the velocity of the cars is given by $\vec{v}_1 = 3t^2\hat{i} - \frac{4}{3}t^6\hat{j}$ and $\vec{v}_2 = 3\left(t^2 - 5t\right)\hat{i}$ respectively. The fictitious (pseudo) force on B as observed by A at t = 1 sec will be :- (m_A = 70 kg, m_B = 50kg)

(1) $70\left(4\hat{j} - 3\hat{i}\right)$

(2) $100\left(4\hat{j} - 3\hat{i}\right)$

(3) $50\left(8\hat{j} - 15\hat{i}\right)$

(4) $70\left(8\hat{j} - 15\hat{i}\right)$

40) A force is applied to two different objects. It produces accelerations of 3 m/s^2 and 6 m/s^2 respectively in these objects. If the same force is applied to the combination of these objects then acceleration of combination will be :-

(1) 2 m/s^2

(2) 4.5 m/s^2

(3) 4 m/s^2

(4) none

41) A 70 kg. man jumps to a height of 0.8 m. Calculate impulse provided by ground to man.

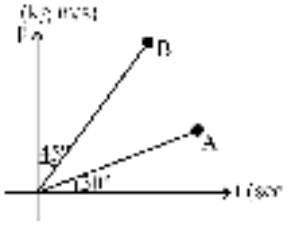
(1) 384.6 kg m/sec

(2) 277.2 kg m/sec

(3) 237.4 kgm/sec

(4) 309.6 kg m/sec

42) For the given p (momentum) - t (time) graph find the ratio of the forces on particles A & B.



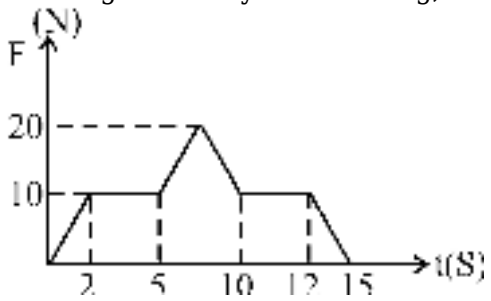
(1) $\sqrt{3} : 1$

(2) $1 : 1$

(3) $\sqrt{3} : 2$

(4) $1 : \sqrt{3}$

43) For the given F-t graph, if force starts acting on a body of mass 2 kg, which is initially at rest, then find its velocity after 15 sec :-



(1) 75 m/s

(2) 150 m/s

(3) 125 m/s

(4) None of these

44) When a force F is applied on a body of mass M, the acceleration produced is a. If three forces equal to F & inclined to each other at 120° act on the same body the acceleration produced will be (all the forces act in the same plane) :-

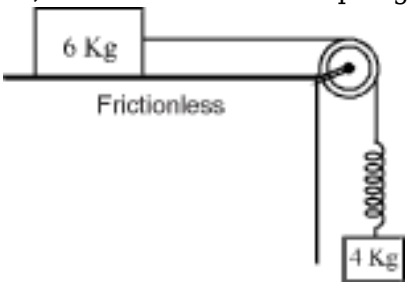
(1) $\sqrt{2}a$

(2) $\sqrt{3}a$

(3) 3a

(4) 0

45) For a system shown in figure, the extension in the spring at steady state is :-



(force constant, K = 240 N/m)

(1) 5 cm

(2) 10 cm

(3) 20 cm

(4) 40 cm

CHEMISTRY

1) Which of the following species can show effective pπ–pπ bonding between the central atom and oxygen ?
(A) CO_2 (B) NO_3^- (C) SiO_2 (D) SO_2

(1) A, B

(2) A, B, D

(3) A, B, C, D

(4) A, C only

2) Which of the following statements are correct regarding NH_3 and PH_3 ?
(A) Both have pyramidal geometry.
(B) Their central atoms have the same hybridisation according to the simplified hybridisation approach.
(C) The bond angle in NH_3 is considerably greater than that in PH_3.
(D) The nature of orbital participation in bonding is identical in both.

(1) A, B

(2) A, C

(3) A, B, C

(4) B, C, D

3) Which statement regarding NH_3 and NF_3 is correct ?

(1) NF_3 has greater dipole moment because N-F bonds are more polar.

(2) NH_3 has greater dipole moment because the resultant bond moments reinforce the contribution arising from the molecular arrangement.

(3) Both have zero dipole moment.

(4) Their dipole moments must be equal because their hybridisation is the same.

4) Which of the following pairs contains species having the same molecular shape but different hybridisation ?

(1) $BeCl_2$ and CO_2

(2) NH_3 and PH_3

(3) I_3^- & N_3^-

(4) BF_3 and SO_3

5) The correct order of N–O bond length in NO_2^-, NO_3^- and NO_2^+ is :

(1) $NO_2^+ < NO_3^- < NO_2^-$

(2) $NO_2^- < NO_3^- < NO_2^+$

(3) $NO_3^- < NO_2^+ < NO_2^-$

(4) $NO_2^+ < NO_2^- < NO_3^-$

6) Which of the following statements are correct for PCl_5 ?
(A) Axial P–Cl bonds are longer than equatorial P–Cl bonds.
(B) Axial bonds experience greater repulsion.
(C) The axial-equatorial angle is 90°.
(D) All five P–Cl bonds are equivalent.

(1) A, B

(2) A, C

(3) A, B, C

(4) B, C, D

7) The correct order of bond angle for NH_3, NF_3, PH_3, PF_3 is :

(1) $NH_3 > NF_3 > PF_3 > PH_3$
(2) $NH_3 > PH_3 > NF_3 > PF_3$
(3) $PF_3 > NF_3 > NH_3 > PH_3$
(4) $PH_3 > PF_3 > NH_3 > NF_3$

8) For a bent molecule XY_2 the resultant dipole moment will be maximum when the angle between the two identical bond moments is :

(1) 120°
(2) 60°
(3) 90°
(4) 134°

9) Which of the following transformations does NOT change the bond order ?

(1) $O_2 \rightarrow O_2^+$
(2) $N_2 \rightarrow N_2^+$
(3) $N_2^+ \rightarrow N_2^-$
(4) $O_2 \rightarrow O_2^-$

10) The force responsible for liquefaction of noble gases is primarily :

(1) Hydrogen bonding
(2) Ion-dipole interaction
(3) Dipole-dipole interaction
(4) Instantaneous dipole-induced dipole interaction

11) Which of the following has the correct boiling-point order ?

(1) $H_2O > HF > NH_3 > CH_4$
(2) $H_2O > NH_3 > HF > CH_4$
(3) $HF > H_2O > NH_3 > CH_4$
(4) $CH_4 > NH_3 > HF > H_2O$

12) Which one of the following combinations contains the incorrect order ?

(1) $NaF < MgF_2 < AlF_3$; Lattice energy
(2) $NaF < MgF_2 < AlF_3$; Covalent character
(3) $NaF < MgF_2 < AlF_3$; Melting point
(4) None of the above

13) Among the following, the number of paramagnetic species is :
B_2, C_2, N_2, O_2, O_2^+, O_2^-

(1) 2

(2) 3

(3) 4

(4) 5

14) **Assertion (A) :** LiF is less soluble in water than other alkali-metal fluorides.
Reason (R) : Its ions have a combination of relatively strong lattice attraction.

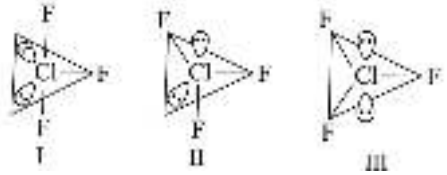
(1) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) **(A)** is incorrect but **(R)** is correct

15) Given below are two statements.
Statement I : For ClF_3 all three possible structures may be drawn as follows.



Statement II : Structure III is most stable, as the orbitals having the lone pairs are axial, where the lp-bp repulsion is minimum.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

16) Match List-I with List-II according to their shape.

	List-I		**List-II**
(A)	XeO_3	(I)	BrF_5
(B)	XeF_2	(II)	NH_3
(C)	XeO_2F_2	(III)	$[I_3]^-$
(D)	$XeOF_4$	(IV)	SF_4

Choose the correct answer from the options given below.

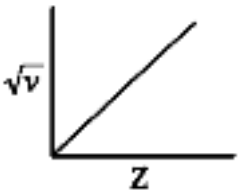
(1) A-II, B-III, C-I, D-IV

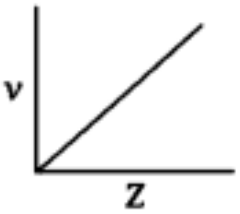
(2) A-II, B-I, C-III, D-IV

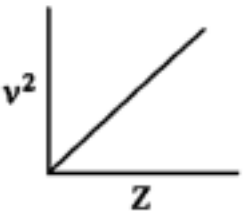
(3) A-II, B-III, C-IV, D-I

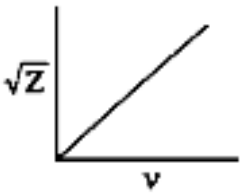
(4) A-III, B-II, C-IV, D-I

17) Henry Moseley studied characteristic X-ray spectra of elements. The graph which represents his observation correctly is :
Given: ν = frequency of X-ray emitted,
Z = atomic number

(1) 

(2) 

(3) 

(4) 

18) Element "E" belongs to period 4 and group 16 of the periodic table. The valence shell electronic configuration of the element, which is just above "E" in the group, is -

(1) $3s^2\ 3p^4$

(2) $3d^{10}\ 4s^2\ 4p^4$

(3) $4d^{10}\ 5s^2\ 5p^4$

(4) $2s^2\ 2p^4$

19) The ionic radii of F^- and O^{2-} respectively are 1.33 Å and 1.4 Å while the covalent radius of N is 0.74 Å. The correct statement for the ionic radius of N^{3-} from the following is :

(1) It is smaller than F^- and N.

(2) It is bigger than O^{2-} and F^-

(3) It is bigger than F^- and N, but smaller than O^{2-}

(4) It is smaller than O^{2-} and F^-, but bigger than N.

20) Match List-I with List-II :

	List-I (Electronic configuration of elements)		**List-II (ΔH_i in kJ mol^{-1})**

(A)	$1s^2\ 2s^2$	(I)	801 kJ mol^{-1}
(B)	$1s^2\ 2s^2\ 2p^4$	(II)	899 kJ mol^{-1}
(C)	$1s^2\ 2s^2\ 2p^3$	(III)	1314 kJ mol^{-1}
(D)	$1s^2\ 2s^2\ 2p^1$	(IV)	1402 kJ mol^{-1}

Choose the correct answer from the options given below :

	A	**B**	**C**	**D**
(1)	II	III	IV	I
(2)	I	IV	III	II
(3)	I	III	IV	II
(4)	IV	I	II	III

(1) 1

(2) 2

(3) 3

(4) 4

21) The incorrect decreasing order of atomic radii is :

(1) Be > Mg > Al > Si

(2) Si > P > Cl > F

(3) Mg > Al > C > O

(4) Al > B > N > F

22) Consider the following ionisation enthalpies (kJ/mol) of two elements 'A' and 'B'.

Element	IE_1	IE_2	IE_3
A	899	1757	17847
B	737	1450	7731

Which of the following statements is correct ?

(1) Both 'A' and 'B' belong to group-2 where 'A' is below 'B'.

(2) Both 'A' and 'B' belong to group-2 where 'B' is below 'A'.

(3) Both 'A' and 'B' belong to group-1 where 'A' is below 'B'.

(4) Both 'A' and 'B' belong to group-1 where 'B' is below 'A'.

23) Which of the following process that is not endothermic in nature ?

(1) $O^-(g) + e^- \rightarrow O^{2-}(g)$

(2) $Na(g) \rightarrow Na^+(g) + e^-$

(3) $H(g) + e^- \rightarrow H^-(g)$

(4) $Ar(g) + e^- \rightarrow Ar^-(g)$

24) **Assertion (A) :** The electron gain enthalpy of halogens becomes continuously less negative from fluorine to iodine.
Reason (R) : Down the group, the size of the atom increases and the attraction between the nucleus and the incoming electron decrease.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

25) The total number of acidic oxides from the following list is :
NO, N_2O, B_2O_3, N_2O_5, CO, SO_3, P_4O_{10}

(1) 3

(2) 4

(3) 5

(4) 6

26) In which of the following arrangements, the sequence is not strictly according to the property written against it ?

(1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing basic character

(2) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(3) HF < HCl < HBr < HI : increasing acid strength

(4) B < C < O < N : increasing Electronegativity

27) What are the shapes and hybridization states of odd-electron species, CH_3 and CF_3 respectively ?

(1) sp^3 and pyramidal, sp^3 and pyramidal

(2) sp^3 and pyramidal, sp^2 and Trigonal planar

(3) sp^2 and Trigonal planar, sp^2 and Trigonal planar

(4) sp^2 and Trigonal planar, sp^3 and pyramidal

28) If the y axis is considered as the Internuclear Axis (INA), which combination of atomic orbitals leads to the formation of a π bond ?

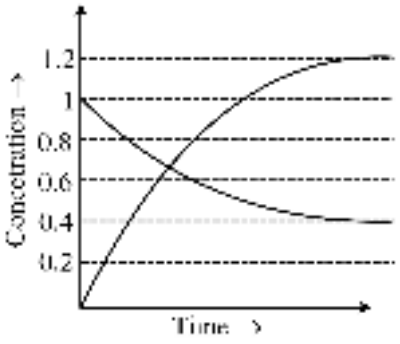
(1) $p_y + p_y$

(2) $p_y + d_{xy}$

(3) $p_x + d_{xy}$

(4) $p_z + d_{xy}$

29) The figure shows the change in concentration of species A and B as a function of time. The



equilibrium constant K_c for the reaction $A(g) \rightleftharpoons 2B(g)$ is :-

(1) $K_c = 3.6$

(2) $K_c = 3$

(3) $K_c = 1.2$

(4) Data insufficient

30)

1 mole of PCl_5 is heated in a 1 litre closed vessel. At equilibrium, 20% moles of PCl_5 are not dissociated. What would be the value of K_C ?

(1) 2.4

(2) 42

(3) 0.32

(4) 3.2

31)

For the reaction :
$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
At equilibrium of above reaction the volume of container was reduced to half. Identify correct statements for new equilibrium.

(1) Moles of SO_3 increase

(2) $[SO_2]$ increases

(3) $[O_2]$ increases

(4) All of the above

32)

At certain temperature A(g) dissociates as $2A(g) \rightleftharpoons 2B(g) + C(g)$

If degree of dissociation is $\alpha(\alpha << 1)$. The expression of K_p, in terms of α and equilibrium pressure P is:-

(1) $\frac{P\alpha^3}{2}$

(2) $\frac{P\alpha^2}{3}$

(3) $\frac{P\alpha^3}{3}$

(4) $\frac{P\alpha^2}{2}$

33)

1 mole N_2 and 3 mole H_2 are placed in a closed container at a pressure of 4 atm. The pressure falls to 3 atm at the same temperature when the following equillibrium is attained

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

The equilibrium constant Kp for dissociation of NH_3 is :-

(1) $\frac{1}{0.5} \times (1.5)^3$ atm^{-2}

(2) $0.5 \times (1.5)^3$ atm^2

(3) $\frac{0.5 \times (1.5)^3}{3 \times 3}$ atm^2

(4) $\frac{3 \times 3}{0.5 \times (1.5)^3}$ atm^{-2}

34)

Following two equilibrium is simultaneously established in a container :-

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$

If some CO(g) is removed from the container, then at new equilibrium :-

(1) PCl_3 concentration will increase

(2) PCl_3 concentration will decrease

(3) $COCl_2$ concentration will remain same

(4) PCl_5 concentration will remain same

35) Two solids dissociate as follows

$A(s) \rightleftharpoons B(g) + C(g)$ $K_{P_1} = x\ atm^2$

$D(s) \rightleftharpoons C(g) + E(g)$ $K_{P_2} = y\ atm^2$

Total pressure when both the solids dissociate simultaneously is

(1) $x^2 + y^2$

(2) $2\sqrt{x+y}$

(3) $x + y$

(4) $\sqrt{x+y}$

36) (i) $N_2O_4 \rightleftharpoons 2NO_2$

(ii) $H_2O_{(\square)} \rightleftharpoons H_2O_{(s)}$

(iii) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(iv) $N_2 + O_2 \rightleftharpoons 2NO$

For above reaction which is/are correct statements.

(a) Increase in pressure does not affect the equilibrium state of (i) & (iv) reaction.
(b) Melting point of ice decreases on increasing pressure.
(c) On increasing temperature at equilibrium all four reaction move to forward direction.
(d) On adding inert gas at constant pressure Ist reaction move to forward direction.

(1) a, b, c
(2) a, b, c, d
(3) b, c, d
(4) b, d

37) The conditions favourable for the reaction
$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$; $\Delta H^{\circ} = -198$ kJ are

(1) Low temperature, high pressure
(2) Any value of T and P
(3) Low temperature and low pressure
(4) High temperature and high pressure

38) $\log\left(\frac{K_p}{K_c}\right) + 2\log RT = 0$ is correct relation for the reaction :-

(1) $PCl_5(g) \rightleftharpoons PC\square_3(g) + C\square_2(g)$
(2) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

39) For the equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$, what is the temperature at which $\frac{K_c(M)}{K_P(atm)} = \frac{1}{3}$?

(1) 0.027 K
(2) 4.6 K
(3) 273 K
(4) 36.54 K

40) $A(s) \rightleftharpoons 2B(g) + C(g)$
solid A is present in a container already containing C at an initial pressure of 0.5 atm. If total pressure at equilibrium is 0.8 atm then find K_P of reaction

(1) 0.012
(2) 0.024
(3) 0.072
(4) 0.036

41) On mixing 10 mL of CCl_4 with 10 mL of benzene, the total volume of the solution is :

(1) > 20 mL

(2) < 20 mL

(3) = 20 mL

(4) cannot be predicted

42) Total vapour pressure of mixture of 1 mol A(P^0_A = 150 torr) and 2 mol B(P^0_B = 240 torr) is 200 torr. In this case :-

(1) there is positive deviation from Raoult's law

(2) there is negative deviation from Raoult's law

(3) there is no deviation from Raoult's law

(4) none of these

43) How much oxygen is dissolved in 100 mL water at 298 K if partial pressure of oxygen is 0.9 atm and $K_H = 1.6 \times 10^4$ atm –

(1) 22.4 mg

(2) 1.1 mg

(3) 2.24 g

(4) 0.01 g

44) For pure water :-

(1) pH increases while pOH decreases with rise in temperature

(2) pH decreases while pOH increases with rise in temperature

(3) both pH and pOH decrease with rise in temperature

(4) both pH and pOH increase with rise in temperature

45) An aqueous solution contains $[H^+] = 10^{-4}$ M. If it is diluted by mixing equal volume of water then the concentration of OH^- in mol dm^{-3} will be :–

(1) $0{\cdot}5 \times 10^{-10}$

(2) 2×10^{-10}

(3) 10^{-6}

(4) 10^{-8}

BIOLOGY

1) In cockroach a ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of :-

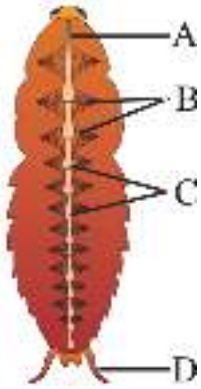
(1) Foregut and Hindgut

(2) Hindgut and midgut

(3) Midgut and foregut

(4) All of the above

2) In the diagram given below, identify the parts labelled as A, B, C & D and select the right option about them.



	Part-A	**Part-B**	**Part-C**	**Part-D**
(1)	Anterior aorta	Heart chambers	Alary muscles	Anal cerci
(2)	Anal cerci	Alary muscles	Heart chambers	Anterior aorta
(3)	Anterior aorta	Alary muscles	Heart chambers	Anal cerci
(4)	Heart chambers	Anal cerci	Anterior aorta	Alary muscles

(1) 1
(2) 2
(3) 3
(4) 4

3) Tergum is joined on the sides by :

(1) Pleuron
(2) Sternum
(3) Alimentary canal
(4) Muscles

4)

The 10th segment bears a pair of jointed filamentous structure called :-

(1) Anal cerci in both sexes
(2) Anal style in both sexes
(3) Anal cerci in male cockroach only
(4) Anal style in male cockroach only

5)

Given below is a list of some structure in frog :-

Metanaphric kidney, Copulatory pads in male, Diaphragm, Amniotic eggs.

How many of the above structures are present in frog.

(1) 1

(2) 2

(3) 3

(4) 4

6) Nymph differs from adult cockroach in –

(1) Absence of wings

(2) Smaller in size

(3) Absence of mature reproductive organs

(4) All of these

7) **Statement-1 :** Ctenophores are commonly known as comb plates.
Statement-2 : The body bears eight external rows of ciliated comb plates, which help in locomotion.

(1) Both statements are correct.

(2) Statement-1 is correct but statement-2 is incorrect.

(3) Statement-1 is incorrect but statement-2 is correct.

(4) Both statements are incorrect.

8) **Statement-1 :** Animals with notochord are called chordate.
Statement-2 : Notochord is a mesodermally derived rod-like structure formed on the ventral side during embryonic development of some animals.

(1) Both statements are correct.

(2) Statement-1 is correct but statement-2 is incorrect.

(3) Statement-1 is incorrect but statement-2 is correct.

(4) Both statements are incorrect.

9) **Statement-1 :** Body of amphibian is divisible into head and trunk.
Statement-2 : Sexes are not separate in amphibians.

(1) Both Statements 1 and 2 are correct.

(2) Both Statements 1 and 2 are incorrect.

(3) Statements-1 is correct but Statement-2 is incorrect.

(4) Statements-2 is correct but Statement-1 is incorrect.

10) Observe the given figure :



Select correct answer for it :

(1) It is fresh water sponge

(2) Animal body is porous

(3) It has radial symmetry

(4) Both 1 & 2



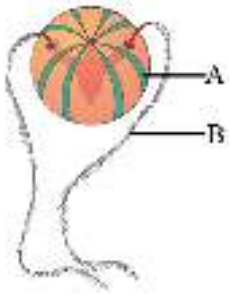
11) Identify the Animal and find out the incorrect option for given animal :

(1) Presence of bioluminescence.

(2) Presence of ten comb plate

(3) Reproduction only sexual

(4) Tissue level organisation



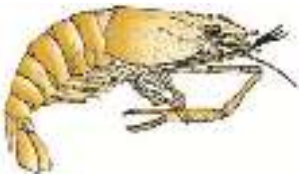
12) Refer the diagrams given below Which of the following option correctly identifies the marked structure and it's function ?

(1) B - flagella - locomotion

(2) B - tentacles - locomotion

(3) A - comb plates - digestion

(4) A - comb plates - locomotion

13) Identify the organism 

(1) Crab

(2) Prawn

(3) Insect

(4) None of these

14) Incomplete digestive tract is found in -

(1) Sponges, Coelenterates, Ctenophores

(2) Coelenterates, Ctenophores, Platyhelminthes

(3) Annelida, Arthropoda, Chordata

(4) Coelenterates, Ctenophores, Echinoderms

15) Select the total number of lizards from the following
Chelone, Calotes, Crocodilus, Hemidactylus, Columba, Neophron

(1) 2
(2) 3
(3) 4
(4) 5

16) In the larval stage of star-fish which type of symmetry is present ?

(1) Radial
(2) Asymmetry
(3) Bilateral symmetry
(4) None of Above

17) **Assertion (A) :** Arthropoda is the largest phylum of kingdom animalia.
Reason (R) : Over two-thirds of all named species on earth are arthropods.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (R) is true but (A) is false

18) **Assertion :** Some of cnidarians shows/exhibit alternation of generation.
Reason : In cnidarians, metagenesis enables species to utilize different reproduction mechanisms according to their environmental conditions.

(1) Assertion and reason are correct but reason is not the correct explanation of assertion.
(2) Assertion and reason both are correct and reason is correct explanation of assertion.
(3) Assertion is correct and reason is incorrect.
(4) Assertion and reason both are incorrect.

19)

Identify the given diagram and choose the correct option :-



(1) Cyclostomata - *Petromyzon* - Jaws present
(2) Chondrichthyes - *Catla* - 6-15 pair of gill slits
(3) Cephalochordata - *Amphioxus* - suctorial mouth without jaws
(4) Cyclostomata - *Petromyzon*, cranium and vertebral column are cartilagenous

20) Which of the following statements are incorrect regarding Aves.

(i) No. of cranial nerves present in birds are 10 pairs.
(ii) Birds are monodelphic
(iii) Hind limbs are modified into wings which help in flying.
(iv) Skull is dicondylic.

(1) Only ii
(2) ii and iii
(3) ii and iv
(4) i, iii and iv

21) Consider following features.
(a) Organ system level of organisation
(b) Multicellular organism.
(c) True coelomates with segmentation of body.
(d) Bilateral symmetry
Select the correct option of animal groups which possess all the above characteristics.

(1) Annelida, Arthropoda and Chordata
(2) Annelida, Arthropoda and Echinodermata
(3) Arthropoda, Mollusca and Chordata
(4) Annelida, Echinodermata and Chordata

22) Read the following statements.
(A) Cephalochordata having notochord through out their life.
(B) Coelenterata and echinodermata have radial symmetry.
(C) Development of true coelom and metamerism developed for first time in annelids.
(D) Air bladder, operculum and viviparity like character present in bony fish.
In light of above statement, choose the correct option which include correct statements :-

(1) A and D only
(2) B and C only
(3) A, B and C only
(4) A, C and D only

23) Radially symmetrical animals exhibiting bioluminescence is characterised by the presence of

(1) Ciliated comb plates
(2) Choanocytes
(3) Ladder like nervous system
(4) Metameric segmentation

24) Identify the correct match from the column-I, II and III.

	Column-I	**Column-II**	**Column-III**
1	Adamsia	a. Sea fur	i. Medusa only
2	Aurelia	b. Sea anemone	ii. Polyp and medusa both

3	Obelia	c. Sea gooseberry	iii. no polyp and medusa
4	Pleurobrachia	d. Jelly fish	iv. Polyp only

(1) 2-b-i; 1-d-iv; 3-a-ii; 4-c-iii

(2) 1-b-i; 2-d-iv; 3-a-ii; 4-c-iii

(3) 1-b-iv; 2-d-i; 3-c-iii; 4-a-ii

(4) 2-d-i; 1-b-iv; 3-a-ii; 4-c-iii



25) Identify the structure (A and B) along with their form.

(1) A-*Aurelia* (Polyp), B-*Adamsia* (Medusa)

(2) A-*Adamsia* (polyp), B-*Aurelia* (Medusa)

(3) A-Adamsia (Medusa), B-Aurelia (Polyp)

(4) A-Aurelia (Medusa), B-Adamisa (Polyp)

26)

Match list-I with list-II :-

List-I		**List-II**	
(A)	Scolidon	(I)	Sting ray
(B)	Pristis	(II)	Dog fish
(C)	Carcharodon	(III)	Saw fish
(D)	Trygon	(IV)	Great white shark

Choose the correct answer from the options given below :-

(1) A-I, B-III, C-IV, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-IV, B-II, C-I, D-III

27) How many of the following characters are shown by phylum Echinodermata:
True coelom, water canal system, evisceration, dorsal heart, open circulatory system.

(1) 3

(2) 4

(3) 2

(4) 5

28) Given below are the diagrams of some animals. find out the suitable statement with related

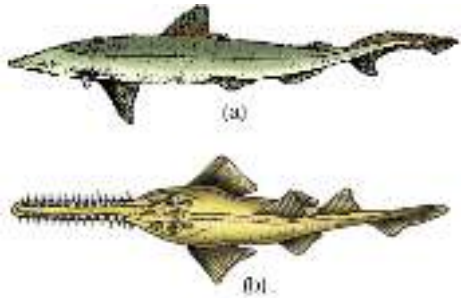
diagram. (i)  (ii)  (iii)  (iv)  (a) Nematocysts present
(b) Flame cells for osmoregulation
(c) Well developed muscular pharynx
(d) Pseudo metamerism

(1) i–a, ii–b, iii–c, iv–d

(2) i–a, ii–c, iii–b, iv–d

(3) i–a, ii–d, iii–b, iv–c

(4) i–d, ii–a, iii–b, iv–c

29)  Identify the statements which are not correct about the diagrams shown above :
(A) Male pectoral fins bears claspers
(B) They show internal fertilization
(C) Air bladder is absent
(D) Teeth are modified cycloid scales

(1) (A), (B)

(2) (A), (D)

(3) (B), (C)

(4) (B), (D)

30) In which one of the following genus name, its two characters and its phylum are not correctly matched, whereas the remaining three are correct ?

	Genus name	**Two characters**	**Class/Phylum**
(1)	Chameleon	→ Three chambered heart → Fertilisation is internal	Reptilia
(2)	Columba	→ Long bones with air cavities → Fore limb convert into wings	Aves
(3)	Macropus	→ Four-chambered heart → Fertilisation is external	Mammalia

(4)	Sea Horse	→ Air bladder present → Two-chambered heart	Fishes

(1) 1

(2) 2

(3) 3

(4) 4

31)

Match list-I with list-II.

	List-I		**List-II**
(a)	Ctenophore example	(I)	Pleurobrachia
(b)	Digestion in ctenophores	(II)	Intra cellular
(c)	Reproduction in ctenophores	(III)	Extra cellular
(d)	Comb plates in ctenophores	(IV)	Both intra & extra cellular
		(V)	Only sexual
		(VI)	Both sexual & Asexual
		(VII)	Ciliated 8 rows
		(VIII)	Flagellated 8 rows

(1) a-I, b-II, c-VI, d-VII

(2) a-I, b-III, c-V, d-VII

(3) a-I, b-IV, c-V, d-VII

(4) a-I, b-III, c-VI, d-VIII

32)

Match column-I with column-II and select correct answer about them.

	Column-I		**Column-II**
a	Hemidactylus	i	Vulture (Scavenger bird)
b	Neophron	ii	Garden lizard
c	Pteropus	iii	Kangaroo
d	Calotes	iv	Wall lizard
e	Macropus	v	Flying fox

(1) a-i, b-ii, c-iii, d-iv, e-v

(2) a-iv, b-i, c-v, d-ii, e-iii

(3) a-iii, b-iv, c-ii, d-i, e-v

(4) a-v, b-iii, c-i, d-iv, e-ii

33) Sinus venosus of frog occurs on

(1) Dorsal side of heart and contain mixed blood
(2) Ventral side of heart and contain deoxygenated blood
(3) Dorsal side of heart and contain deoxygenated blood
(4) Ventral side of heart and contain mixed blood

34)

Which of the statement is wrong about frog ?

(1) The skin is smooth and slippery due to presence of mucus
(2) The colour of dorsal side of body is pale yellow and ventral side is olive green
(3) The colour dorsal side of body is olive green and ventral side is pale yellow
(4) The most common species of frog in India is *Rana tigrina*

35) Read the following statement w.r.t frogs :
(i) They are cold blooded
(ii) They perform camouflage to hide from their enemies
(iii) They undergo hibernation but no aestivation
(iv) The frog drinks water
(v) The skin usually remains dry
How many of the above statements are correct ?

(1) Four
(2) Two
(3) Five
(4) Three

36) Choose the incorrect statement for frog

(1) Body of frog divisible into Head and Trunk
(2) Tympanum receives sound signals
(3) Hind limbs bear four digits with web
(4) Fore limbs bear four digits

37) Choose the incorrect statement with respect to frog :-

(1) Vocal sacs are found only in male
(2) Copulatory pads are found in forelimbs of male frog
(3) Neck is absent
(4) Hindlimbs have four digits and forelimbs have five digits

38) About frog, select incorrect statement only

(1) The skin is smooth and slippery
(2) Dorsal side of body is generally olive green

(3) On the ventral side the skin is pale yellow

(4) Frog drink water but absorb it through skin

39) Which statement is not correct about *Rana Tigrina* ?

(1) They do not have constant body temperature

(2) They have the ability to change the colour

(3) They can live only on land and not in fresh water

(4) They shows the protective colouration mimicry

40) How many statement are correct about Frogs.
(i) Feet have webbed digits that helps in flying.
(ii) Male frogs can be distinguished by presence of sound producing vocal sacs.
(iii) The hind limb end in five digits.
(iv) On either side of eyes a membranous tympanum receives sound signals.

(1) One

(2) Two

(3) Three

(4) Four

41) Which statement is correct about digestive system of frog ?

(1) Digestive system consists of only alimentary canal.

(2) Alimentary canal is long.

(3) Liver secretes the bile juice that stored in pancreas.

(4) Partially digested food is called chyme is passed from stomach to first part of small intestine.

42) Which statement is incorrect about excretion of frog ?

(1) The excretory system consists of a pair of kidneys.

(2) Kidney of frog are compact dark red and bean like structure.

(3) Kidney has several functional unit called nephrons.

(4) Kidneys contain Bidder's canal to transport sperms and urine.

43) Which of the following statement is correct related to the excretory system of frog ?

(1) Two ureters emerge from the each kidney

(2) Ureter and oviduct combine to form the urogenital duct in female

(3) Frog excretes uric acid hence uricotelic

(4) Excretory waste is separated and excreted by the kidney

44) Choose the incorrect statement about frog -

(1) Frog is carnivores.

(2) Alimentary canal is too long.

(3) Skin acts as aquatic respiratory organ.

(4) During hibernation and aestivation gaseous exchange takes place through skin.

45) Choose correct statements ?

(a) The most common species of the frog found in India is *Rana tigrina*.

(b) Dorsal side of frog is olive green.

(c) Compulatory pad and vocal sacs are found in both male and female.

(d) Winter sleep is known as aestivation.

(1) (a) & (b)

(2) (a) & (c)

(3) Only (a)

(4) (a) & (d)

46) Choose incorrect statement ?

(1) Frogs have three chambered heart and are neurogenic in nature.

(2) Main excretory product of frog is urea.

(3) Their system for control and coordination is highly evolved and includes both neural system and endocrine glands.

(4) RBC of frog are nucleated and contain red coloured pigments called haemoglobin.

47) Identify the organism. 

(1) Sycon

(2) Euspongia

(3) Aurelia

(4) Spongilla

48) Which one of the following is correct ? (A)  (B)  (C) 

(D) 

(1) All these animals are aquatic, free living.

(2) All are true coelomates.

(3) "A" has radial symmetry but remaining have bilateral symmetry.

(4) A is monoecious but remaining are dioecious.

49) **Statement-I :** Bones have hard & pliable ground substance.
Statement-II : Cartilage has solid & non-pliable matrix.

(1) Statement I & II both are correct.

(2) Statement I is correct & II is incorrect

(3) Statement I is incorrect & II is correct

(4) Both statement I & II is incorrect

50) Find out correct option with respect to given statements-
Statement-I : Cells of connective tissue secrete some polysaccharide materials.
Statement-II : The ground substance of connective tissue is mainly a protein product synthesized by cells of connective tissue.

(1) Both statement I & II are correct

(2) Both statement I & II are incorrect

(3) Statement I is correct and statement II is incorrect

(4) Statement I is incorrect and statement II is correct

51) Given below are two statements :–
Statement-I : Connective tissue are most abundant and widely distributed in body of simple animals only.
Statement-II : They are named connective tissue because of their special function of linking and supporting other tissue/organ of the body.
Correct answer from the options given below :-

(1) Both statement I and statement II are false.

(2) Statement I is correct but statement II is false.

(3) Statement I is incorrect but statement II is true.

(4) Both statement I and statement II are true.

52) Which type of tissue correctly matches with its location ?

	Tissue	**Location**
1	Smooth muscle	Wall of intestine
2	Areolar tissue	Tendons
3	Transitional epithelium	Tip of nose
4	Cuboidal epithelium	Lining of stomach

(1) 1

(2) 2

(3) 3

(4) 4

53) Match the following column :

	Column-I		**Column-II**

(A)	Unicellular Gland	(i)	Goblet cell
(B)	Tight Junction	(ii)	For rapid transfer of ions small molecules
(C)	Gap Junction	(iii)	Perform cementing to keep neighbouring cells together
(D)	Adhering junction	(iv)	Help to stop substances from leaking across tissue

(1) A-i, B-iv, C-ii, D-iii

(2) A-i, B-ii, C-iii, D-iv

(3) A-ii, B-iii, C-i, D-iv

(4) A-iii, B-i, C-ii, D-iv

54)

Read the following statement and choose the correct one regarding animal tissue.

(1) Few complex animals consist of only four basic types of tissue.

(2) All complex animals consist of only four basic types of tissue.

(3) Most of the complex animals consist of only four basic types of tissue.

(4) All complex animals consist of some basic types of tissue.

55)

Match column-I with column-II with respect to connective tissue -

	Column-I		**Column-II**
(i)	Tendon	(a)	Dense regular
(ii)	In the skin	(b)	Dense irregular
(iii)	Fat storage	(c)	Adipose C.T
(iv)	Support framework for epithelium	(d)	Areolar connection tissue

(1) i-a, ii-b, iii-c, iv-d

(2) i-b, ii-c, iii-d, iv-a

(3) i-c, ii-d, iii-a, iv-b

(4) i-a, ii-c, iii-b, iv-d

56) Hollow woolen ball like shows which type of structure?

(1) Primary

(2) Secondary

(3) Tertiary

(4) Quaternary

57) Match the **column I** with **column II** and choose the correct combination from the options given.

	Column I (Component)		Column II (% of the total cellular mass)
(A)	Lipids	(i)	1
(B)	Ions	(ii)	2
(C)	Carbohydrates	(iii)	3
(D)	Nucleic acids	(iv)	5-7

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(ii), B-(i), C-(iii), D-(iv)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(iv), B-(ii), C-(iii), D-(i)

58) Identify the **incorrect** statement from following options :-

(1) Rubber and Gums are secondary metabolites.

(2) Heterocyclic compound in nucleic acid are the nitrogenous bases.

(3) Cellulose and Glycogen are heteropolymer.

(4) Most abundant element in human body is oxygen.

59) **Assertion :** Haemoglobin is the example of quaternary structure of protein.
Reason : Haemoglobin consist of 4-subunits (two α-types and two β-types)

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

60) A competitive inhibitor of succinic dehydrogenase is:

(1) Malate

(2) Malonate

(3) Oxaloacetate

(4) α-ketoglutarate

61) Which is/are **correct** ?

(1) An enzyme like any protein has primary, secondary and tertiary structure.

(2) Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80 - 90°C)

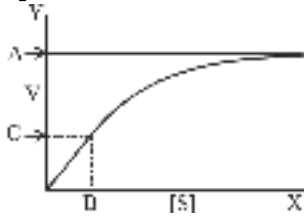
(3) Ribozymes are nucleic acids behaving like enzyme

(4) All are correct

62) **Assertion:** Biocatalyst differ from inorganic catalysts.
Reason: Unlike inorganic catalysts biocatalysts damaged at high temperature.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

63) The adjoining graph shows effect of change in concentration of substrate on enzyme activity.



Identify **A**, **B** and **C**.

	A	**B**	**C**
(1)	$\frac{V_{max}}{2}$	Km	V_{max}
(2)	$\frac{V_{max}}{2}$	Km	Substrate concentration
(3)	V_{max}	Km	$\frac{V_{max}}{2}$
(4)	Km	V_{max}	$\frac{V_{max}}{2}$

(1) 1
(2) 2
(3) 3
(4) 4

64) Who provided evidence for production of glucose when plants grow ?

(1) Julius von sachs
(2) Joseph Priestley
(3) Joan Ingenhousz
(4) Van Niel

65) Stroma lamellae membrane has :-

(1) P.S.-II only
(2) P.S.-I only
(3) P.S.-I and NADP reductase enzyme
(4) P.S.-I and water splitting complex

66) ATP : CO_2 consumption ratio during the photosynthesis in the C_3 plant :

(1) 3 : 1
(2) 1 : 3
(3) 2 : 1

(4) 1 : 4

67) The first stable product of CO_2 fixation in *Sorghum* in bundle sheath cell is

(1) Phosphoglyceric acid
(2) Pyruvic acid
(3) Oxaloacetic acid
(4) Succinic acid

68) Which statement is **incorrect** ?

(1) C_3 plants respond to higher temperature, show higher photosynthesis rate while C_4 plants have lower temperature optimum.
(2) Tropical plants have higher temperature optimum than the plants adapted to temperate climate.
(3) Light reaction is less temperature sensitive than dark reaction.
(4) The effect of water as a factor is more through its effect on plant rather than directly on photosynthesis.

69) **Assertion (A) :** In C_4 plants photorespiration does not occur.
Reason (R) : C_3 acid from the mesophyll is broken down in the bundle sheath cells to release CO_2 that increases the concentration of CO_2 at the RuBisCO enzyme site.
Choose the correct option :

(1) Both A and R are correct and R is correct explanation of A
(2) Both A are R are correct but R is not correct explanation of A
(3) A is correct, R is incorrect
(4) Both A and R are incorrect

70) Given below are two statements :
Statement-I : Photosynthesis does take place in the green leaves of plants but it does so in other green parts of the plants.
Statement-II : Usually the chloroplast align themselves along the walls of the mesophyll cells, such that they get the optimum quantity of incident light.
Choose the correct option :

(1) Statement I and statement II both are correct
(2) Statement I and statement II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

71) Read the following passage and choose the appropriate answer-
Plants that are adapted to dry tropical regions have the C_4 pathway.
Though these plants have the C_4 oxaloacetic acid as the first CO_2 fixation product they use the C_3 pathway or the Calvin cycle as the main biosynthetic pathway.
C_4 plants are special: They have a special type of leaf anatomy, they tolerate higher temperatures, they show a response to high light intensities, they lack a process called photorespiration and have greater productivity of biomass.
Statement I: The bundle sheath cells may form several layers around the vascular bundles; they are

characterised by having a large number of chloroplasts, thick walls impervious to gaseous exchange and large intercellular spaces.
Statement II: In the C_4 plants Calvin cycle does not take place in the mesophyll cells but does so only in the bundle sheath cells.

(1) Only statement I is correct

(2) Both statements are incorrect

(3) Both statements are correct

(4) Only statement I is incorrect

72) Consider the following statements
(i) Green plant carry out photosynthesis a physico-chemical process by which they use light energy to drive the synthesis of organic compounds
(ii) Photosynthesis is responsible for the release of oxygen into the atmosphere by green plants
(iii) The green plants make or rather synthesise the food they need through photosynthesis and are therefore called heterotrophs.
(iv) There is a clear division of labour within the chloroplast
(v) The membrane system of chloroplast is responsible for trapping the light energy and also for the syntheis of ATP and glucose
Choose the correct answer from the options given below :

(1) i, ii, iii are true

(2) ii, iii, iv are true

(3) iii, iv, v are true

(4) i, ii, iv are true

73) Which of the following step/s during glycolysis is associated with utilization of ATP ?

(1) Glucose → Glucose – 6 – phosphate

(2) Fructose-6-phosphate → Fructose-1,6-bisphosphate

(3) PEP → Pyruvic acid

(4) Both (1) and (2)

74) In TCA cycle, isomerisation of citrate is followed by how many decarboxylations to form succinyl CoA?

(1) Two

(2) Three

(3) One

(4) Four

75) Which of the following is known as complex I of mitochondrial E.T.S ?

(1) NADH dehydrogenase

(2) Cytochrome bc_1 complex

(3) Cytochrome c oxidase

(4) Cytochrome c

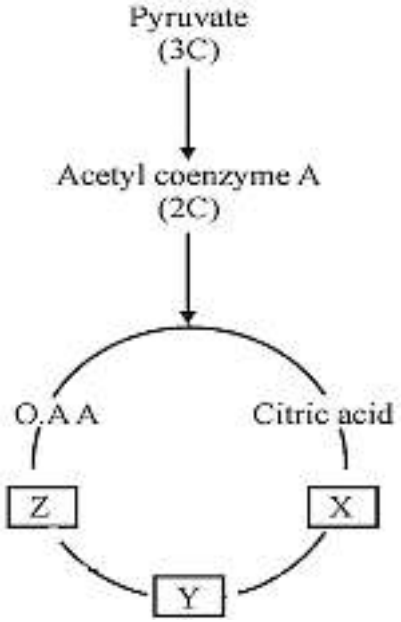
76) Acetyl Co-A can be produced by :-

(1) Pyruvic acid
(2) Fatty acid
(3) Protein
(4) All of these

77) **Assertion :** TCA cycle starts with the condensation of acetyl group with OAA & water to yield citric acid.
Reason : TCA cycle occurs in the mitochondria.

(1) Both Assertion & Reason are True & the Reason is the correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

78) The given figure is of citric acid cycle-



	X	Y	Z
(1)	α-Ketoglutaric acid	Succinic acid	Malic acid
(2)	Succinic acid	α-Ketoglutaric acid	Malic acid
(3)	Succinic acid	Malic acid	α-Ketoglutaric acid
(4)	Malic acid	α-Ketoglutaric acid	Succinic acid

(1) 1
(2) 2
(3) 3
(4) 4

79) Identify the **incorrect** statement :

(1) The compound that are oxidised during respiration are known as respiratory substrates

(2) Breakdown of glucose to pyruvic acid is called glycolysis

(3) Sucrose is converted into glucose and fructose by the enzyme invertase

(4) During the conversion of one molecule succinyl CoA to succinic acid, two molecules of GTP are synthesised

80) In the Glycolysis process glucose undergoes oxidation to form two molecules of pyruvic acid. This process occurs in the cytoplasm of the cell.
Given below are two statements about this paragraph-
Statement-I : During the process of glycolysis, glucose undergoes complete oxidation to form pyruvic acid.
Statement-II : In plants the glucose required for this process is primarily derived from sucrose or storage carbohydrates.
Choose the correct answer from the options given below-

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is incorrect but statement-II is correct

(4) Statement-I is correct but statement-II is incorrect

81) Read the following statements & select the correct ones.
a. Cellular respiration is a redox reaction in which food molecules get oxidised and coenzymes are reduced
b. Energy released in oxidation of substrates is trapped is CO_2 which is used for various activity of cell.
c. The role of oxygen is limited to terminal step of respiration.
d. During alcoholic fermentation glucose is partially oxidised by using O_2 & release $2CO_2$ only.
e. Respiration in mitochondria occurs in the presence of oxygen.

(1) a, b, d correct

(2) a, b, e correct

(3) a, c, e correct

(4) b, d, e correct

82) Fruit ripening hormone is :-

(1) Auxin

(2) Cytokinin

(3) Gibberellic acid

(4) Ethylene

83) Which of the following plant growth regulators play an important role in increasing tolerance of plant to various kinds of stresses?

(1) Auxin

(2) Gibberellin

(3) ABA

(4) Ethylene

84) Who confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas.

(1) F. W. Went
(2) H. H. Cousins
(3) F. Skoog
(4) Miller

85) **Assertion (A) :-** During arithmetic growth, there is growth at constant rate.
Reason (R) :- In arithmetic growth, of the two only one daughter cell continues to divide while the other differentiates and matures.

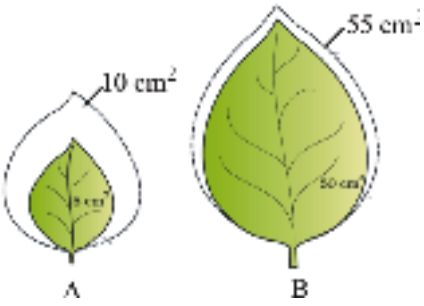
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

86) Which of the following statements is **incorrect** ?

(1) The phenomenon of heterophylly is an example of plasticity.
(2) Broadly, development is considered as the sum of growth and differentiation
(3) Cytokinins initiate rooting in stem cuttings,
(4) The PGR ,ABA play an important role in plant response to wounds

87)

Two leaves A & B with their growth are depicted below:-



Which of the following statements is **incorrect** regarding above situation ?

(1) Absolute growth is same in A and B
(2) Relative growth of A is twice as that of B
(3) Relative growth of B is different from relative growth of A
(4) Relative Growth of B is 1/10 of relative growth of A

88) Which of the following plant growth regulator, is derivative of accessory photosynthetic pigments ?

(1) Abscisic acid
(2) Gibberellic acid
(3) Indole 3 - acetic acid

(4) Ethylene

89) Which of the following statement is/are **incorrect** ?
a) Auxins can promote parthenocarpy in some plants like tomatoes.
b) NAA (Napthalene Acetic Acid) and 2,4-D are natural auxins commonly used in agriculture.
c) Gibberellins promote bolting in rosette plants.
d) Cytokinins are synthesized in roots and they promote nutrient mobilization.
e) ABA (Abscisic Acid) promotes fruit ripening and is also known as the stress hormone.

(1) (a) and (c) only
(2) (b), (c) and (e)
(3) (a), (c), and (d)
(4) (b) and (e)

90)

How many of the following functions/effects are related to the cytokinins ?

bolting, lateral shoot growth, overcome apical dominance, apical hook formation, adventitious shoot formation, delay of leaf senscence, breaking seed dormancy.
(1) Four
(2) One
(3) Two
(4) Three

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	1	3	2	1	3	2	3	2	1	4	4	2	4	1	4	4	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	1	3	1	2	1	1	1	3	1	2	3	2	1	3	1	1	2	1
Q.	41	42	43	44	45															
A.	2	4	1	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	2	3	4	3	1	2	3	4	1	4	3	1	1	3	1	1	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	3	4	2	2	4	3	1	4	4	1	2	2	2	4	1	4	4	2
Q.	86	87	88	89	90															
A.	1	2	4	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	1	1	1	4	3	2	3	4	2	4	2	2	1	3	1	2	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	1	4	2	2	1	1	2	3	3	2	3	2	2	3	4	4	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	4	2	1	1	2	4	4	3	3	1	1	2	1	3	2	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	3	1	2	1	1	1	3	1	4	4	4	1	1	4	2	1	4	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	4	3	2	1	3	2	1	4	1										

SOLUTIONS

PHYSICS

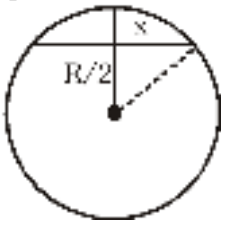
1) $W = q\Delta V$

$= q(V_\infty - V_C)$

$= q\left[0 - 3\frac{kq}{(a/\sqrt{3})}\right] = \frac{-3\sqrt{3}kq^2}{a}$

2) $\phi = \frac{q_{in}}{\in_0}$

$q = \lambda \cdot \ell$



and length

$\ell = 2x = 2\sqrt{R^2 - \frac{R^2}{4}} = \sqrt{3}R$

$\phi = \frac{\sqrt{3}\lambda.R}{\in_0}$

3) Entire charge of inner sphere will flow to outer shell so that their potential is same.

4)

At α-particle, E = field only due to deutron.

$E = \frac{Ke}{r^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{10^{-20}} = 1.44 \times 10^{11}$ N/C

5) **Question Explanation:** A total charge of 1 pC is uniformly distributed on a ring. We need to find the electric potential at the centre of the ring.

Concept: Electric Potential due to Charged Ring

Solution: For a uniformly charged ring, the electric potential at the centre is: $V = \frac{kQ}{R}$

$Q = 1$ pC $= 1 \times 10^{-12}$ C

Diameter of the ring = 1 mm

$R = \frac{1}{2}mm = 0.5mm = 5 \times 10^{-4}m$

$k = 9 \times 10^9$ N m^2 /C^2

$V = \frac{9 \times 10^9 \times 1 \times 10^{-12}}{5 \times 10^{-4}} = \frac{9 \times 10^{-3}}{5 \times 10^{-4}} = 18$ V

Final answer: option (2)

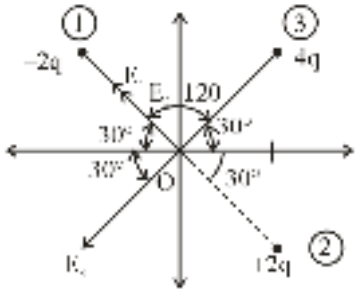
6)

By come

$K_1 + U_1 = K_2 + U_2$

$$0 + \frac{Ke^2}{r} = 2\left(\frac{1}{2}m_\rho v^2\right) + 0$$

$$v = e\sqrt{\frac{K}{m_\rho r}}$$



7)

At origin

field due to –2q & + 2q

$E = E_1 + E_2$

$$E = \frac{4Kq}{r^2} \quad ...(1)$$

Due to 4q $E_3 = \frac{4Kq}{r^2} \Rightarrow E_R = \sqrt{E^2 + E_3^2 + 2E_1E_3\cos 60^o}$

$$= \frac{4Kq}{r^2}\sqrt{3}$$

$$E_R = \frac{4\sqrt{3}Kq}{r^2}$$

8)

Explaining the Question :- Asking about the curved surface B, if point charge "q is placed at the center of circular surface 'c'

Concept:- This question is based on Gaissi law

Solution :-

Total flux

$$\phi_T = \frac{q}{2\varepsilon_0}$$

$$\phi_T = \phi_A + \phi_B = \frac{q}{2\varepsilon_0}$$

$$\phi_B = \frac{q}{2\varepsilon_0} - \phi$$

Final Answer :- 2

9)

$V_1 = 16 = \frac{KQ}{r_1}$, $v_2 = \frac{KQ}{r_2}$ (Q - Remains same)

$V_2 = V_1 \left(\frac{r_1}{r_2}\right) = 16 \left(\frac{r}{2r}\right) = 8V$

10)

Question Explanation:
A charged ring of radius R has charge +q. A test charge $-q_0$ is released from rest on the axis of the ring at a distance $\sqrt{3}R$ from the center. We are to find the kinetic energy the test charge gains when it reaches the center of the ring.

Concept: Conservation of Energy and Electrostatic Potential

Solution:
Kinetic energy gained decrease in electric potential energy:

$$\Delta K = -\Delta U = q_0 (V_i - V_f)$$

Electric potential on the axis of a uniformly charged ring:

$$V(x) = \frac{1}{4\pi\varepsilon_0} \frac{q}{\sqrt{R^2 + x^2}}$$

Initial position: $x = \sqrt{3}R$

$$V_i = \frac{1}{4\pi\varepsilon_0 \sqrt{R^2 + 3R^2}} = \frac{1}{4\pi\varepsilon_0} \frac{q}{2R}$$

Final position (centre): x = 0

$$V_f = \frac{1}{4\pi\varepsilon_0} \frac{q}{R}$$

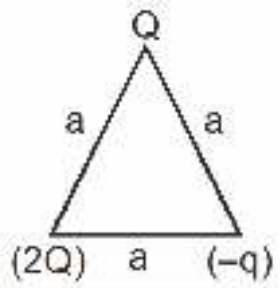
Kinetic energy gained:

$$K = q_0(V_f - V_i) = q_0 \left(\frac{1}{4\pi\varepsilon_0}\right) \left(\frac{q}{R} - \frac{q}{2R}\right)$$

$$K = \frac{1}{4\pi\varepsilon_0} \frac{qq_0}{2R}$$

Final answer: option (2)

11) **Question Explanation :**



If total electric potential energy in positive, then we have to write relation between 'Q' and 'q'.

Concept: Potential energy.

Formula: $U_{q_1 q_2} = \frac{kq_1 q_2}{r}$

Solution:

$$U_{system} = \frac{kQ(2Q)}{a} + \frac{k(2Q)(-q)}{a} + \frac{k(-q)(Q)}{a}$$

$$U_{system} = \frac{2kQ^2}{a} - \frac{3kQq}{a}$$

Given $U_{system} > 0$

$$\frac{2kQ^2}{a} - \frac{3kQq}{a} > 0$$

$$\frac{2kQ^2}{a} > \frac{3kQq}{a}$$

$2Q > 3q$

Final Answer: Option (1)

12)

After redistribution - $q_1 = q_2 = = \frac{10+2}{2} = 6\mu C$

Charge transferred from A → $\Delta q = 10 - 6 = 4\mu C$

⇒ mass transferred = $\Delta m = m_e\left(\frac{\Delta Q}{e}\right)$

$$\Delta m = \frac{9.1 \times 10^{-31} \times 4 \times 10^{-6}}{1.6 \times 10^{-19}} = 2.2 \times 10^{-17}\ kg$$

13) $B = \mu^0 \mu_r \left(\frac{N}{2\pi r}\right) i$

$$1 = 4\pi \times 10^{-7} \times \mu_r \times \frac{400}{40 \times 10^{-2}} \times 2$$

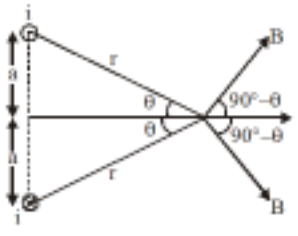
$$\mu_r = \frac{10^4}{8\pi} \simeq 400$$

14) $E = -(\vec{V} \times \vec{B})$

$$E(-\hat{j}) = -(V\hat{i} \times \vec{B})$$

$$E\hat{j} = V\hat{i} \times \vec{B}$$

$\Rightarrow \vec{B} = B(-\hat{k})$

15) 

Angle between MF produced by both wires are (180° – 2θ) so net MF at P will be

$B_P = \sqrt{B^2 + B^2 + 2BB\cos(\pi - 2\theta)}$

$= 2B \sin\theta$

$= 2\left(\frac{\mu_0 i}{2\pi r}\right)\frac{a}{r}$

$= \frac{\mu_0 i a}{\pi r^2}$

16) loop = 1 $\oint B \times d\ell = \mu_0\left(-i + \hat{i} - i\right) = -\mu_0 i$

loop = 2 $\oint B \times d\ell = \mu_0(+i - i + i) = \mu_0 i$

loop = 3 $\oint B \times d\ell = \mu_0(-i - i + i) = -\mu_0 i$

loop = 4 $\oint B \times d\ell = \mu_0(i + i - i) = \mu_0 i$

17) **Explain Question:**

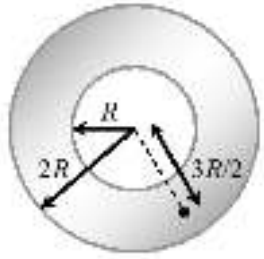
A hollow cylindrical conductor of inner radius R and outer radius 2R carries a uniformly distributed current i along its axis. Find the magnetic induction at point P located at a distance $\frac{3R}{2}$ from the axis.

Concept:

Magnetic field due to a current-carrying hollow conductor (Ampere's law)

Solution:

Since R < $\frac{3R}{2}$ < 2R, point P lies within the conducting material.



Current density

$J = i / [\pi((2R)^2 - R^2)] = i / (3\pi R^2)$

Current enclosed within radius r = $\frac{3R}{2}$

$I_{enclosed} = J \times \pi (r^2 - R^2)$

$$= \frac{i}{3\pi R^2} \times \pi \left(\left(\frac{9R^2}{4} \right) - R^2 \right) = \left(\frac{i}{3} \right) \times \left(\frac{5}{4} \right) = \frac{5i}{12}$$

Using Ampere's

$B \times 2\pi r = \mu_0 I_{enclosed}$

$B = \mu_0 (5i / 12) / [2\pi \times (3R / 2)]$

$= 5\mu_0 i / (36\pi R)$

Final answer: Option (4)

18)

Explain Question : Find the magnetic field at the centroid O of an equilateral triangle made of uniform wires AB, BC, CA.
When current I enters at vertex A and leaves at the midpoint of BC.

Concept : This question is based on Use Biot-Savart plus geometric symmetry : Mirror symmetry about the median from A to midpoint of BC determines how fields from symmetric current branches combine.

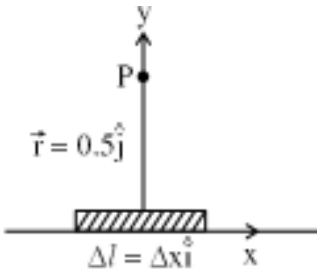
Formula : Biot-Savart : $dB = \frac{\mu_0}{4\pi} \frac{Id\ell \times \hat{r}}{r^2}$. Integrate along each wire, use symmetry to compare contributions.

Solution :

Current splits equally into AB and AC. The configuration is symmetric about the median through A; corresponding current elements on AB and AC produce equal-magnitude fields at O but in opposite directions (by mirror reflection). Their sum (including contributions from BC's two equal halves directed toward the midpoint) cancels, yielding $\vec{B} = 0$.

Final answer : Option (4)

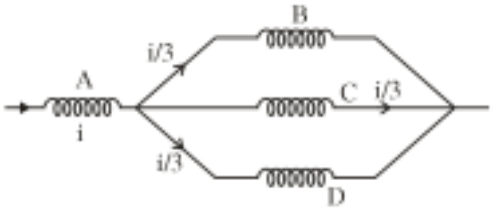
19)



$$\vec{dB} = \frac{\mu_0}{4\pi} \frac{I(\vec{dl} \times \vec{r})}{r^3} \text{ (Tesla)}$$

$$= \frac{10^{-7} \times 10 \times \left(\frac{1}{2} \times \frac{1}{100} \right) (+\hat{k})}{\left(\frac{1}{2} \right)^3}$$

$$= 4 \times 10^{-8} T\ (+\hat{k})$$



20)

$\phi \propto i$

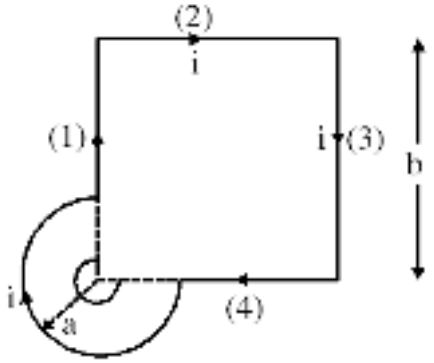
$\Rightarrow B \propto i$

so, field at centre of C = $\frac{3}{3} = 1T$

21)

$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$ if x >> R

$B \cong \frac{\mu_0 I R^2}{2x^3}$



22)

Magnetic field due to wire (1) & wire (4) is zero

$B_2 = B_3 = \frac{\mu_0}{4\pi}\frac{i}{b}\left(\sin\frac{\pi}{4} + \sin\theta\right)$

$= \frac{\mu_0}{4\pi}.\frac{i}{b} \times \frac{1}{\sqrt{2}} \otimes$

$B_{arc} = \frac{\mu_0}{4\pi}\frac{i}{a}\left(270 \times \frac{\pi}{180}\right)$

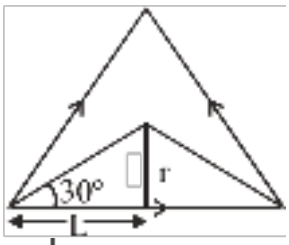
$= \frac{\mu_0}{4\pi}\frac{i}{a} \times \frac{3\pi}{2}$

$B_{at\ O} = B_{arc\ C} + B_2 + B_3$

$= \frac{\mu_0}{4\pi} \times \frac{i}{a} \times \frac{3\pi}{2} + 2 \times \frac{\mu_0}{4\pi}.\frac{i}{b} \times \frac{1}{\sqrt{2}}$

$= \frac{\mu_0}{4\pi}.i\left(\frac{3\pi}{2a} + \frac{\sqrt{2}}{b}\right)$

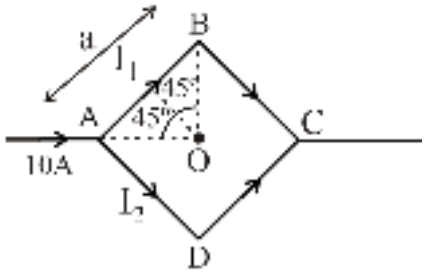
23) $\frac{r}{L} = \tan 30°$



$r = \frac{L}{\sqrt{3}}$

$B = 3\left[\frac{\mu_0 i}{4\pi r}(\sin 60^\circ + \sin 60^\circ)\right] = \frac{3\mu_0 i}{4\pi \frac{L}{\sqrt{3}}}\left[\sqrt{3}\right] = \frac{9}{4}\frac{\mu_0 i}{\pi L}$

24) $r = \sqrt{\frac{2mk}{qB}}$



25)

$I_1 = \frac{10 \times 2r}{2r + r} = \frac{20}{3}A$

$I_2 = \frac{10}{3}A$

$\vec{B}_{AB} = \vec{B}_{BC} = \frac{\mu_0 I_1}{4\pi\frac{a}{2}} \times 2\sin 45 \otimes$

$= \frac{\mu_0 I_1}{\sqrt{2}\pi a} \otimes$

Similarily : $\vec{B}_{AD} = \vec{B}_{DC} = \frac{\mu_0 I_2}{\sqrt{2}\pi a} \odot$

$B_{Net} = \frac{2\mu_0}{\sqrt{2}\pi a}(I_1 - I_2) \otimes$

$= \frac{10\sqrt{2}\mu_0}{3\pi a} \otimes$

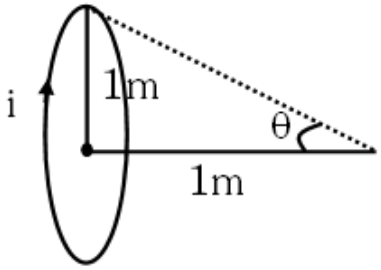
26)

By ampere law

$\oint \vec{B}_{in}.\vec{dl} = 0$, $B_{in} = 0$

$\oint \vec{B}_{out}.\vec{dl} = \mu_0 I$, $B_{out} \neq 0$

Magnetic field will be only outside the pipe.



27)

$B_A = B_0 \sin^3 \theta$

$= \frac{\mu_0 i}{2R} \times \sin^3\theta$

$= \frac{4\pi \times 10^{-7} \times \sqrt{2}}{2 \times 1} \times \left(\frac{1}{2\sqrt{2}}\right)$

$B_A = 3.14 \times 10^{-7}$ T

28) $\frac{q_H}{q_\alpha} = \frac{1}{2}$

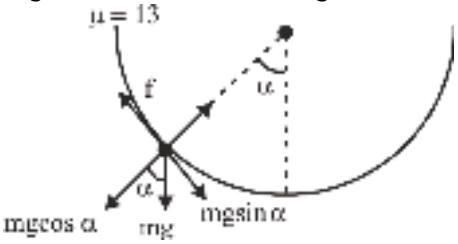
$r = \frac{mv}{qB}$

For same momenta, $r \propto \frac{1}{q}$

$\frac{r_H}{r_\alpha} = \frac{q_\alpha}{q_H} = \frac{2}{1}$

29)

The two forces acting at the insect are mg and N. Let us resolve mg into two components. mg cos α balances N mg sin α is balanced by the frictional force.



 N = mg cosα

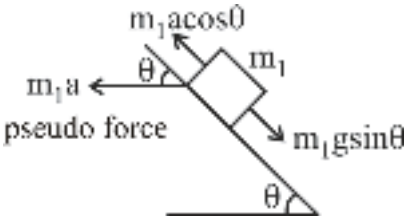
f = mg sinα, But f=μN=μ mg cosα

∴ μmg cos α = mg sinα ⇒ cotα = $\frac{1}{\mu}$ ⇒ cotα =3

30) If no relative motion, both blocks move horizontally with acceleration 'a'.

then $F = (m_1 + m_2)a$ ...(1)

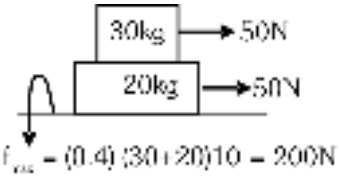
for m_1



If m_1 is at rest w.r.t. m_2 , then component of pseudo force m_1a along incline is $m_1a \cos\theta$ which

balances $m_1 g\sin\theta$ so, $m_1 a\cos\theta = m_1 g\sin\theta$
$a = g\tan\theta$ ...(2)
put in (1)
$F = (m_1 + m_2)\, g\tan\theta$

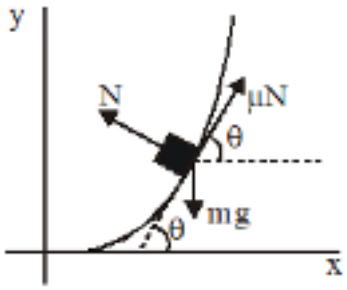
31) $\mu = \tan\theta\left(1 - \frac{1}{n^2}\right) = \tan\theta\left(1 - \frac{1}{2^2}\right) = \frac{3}{4}\tan\theta$



32)
$a_B = 0$
$a_A = \frac{50}{30} = \frac{5}{3}\,m/s^2$

33) Reading of weighing scale $= m(g + a)$
$= 80(10+5)=1200$ N

34) $y = \frac{x^2}{20}$ $\Rightarrow \frac{dy}{dx} = \frac{x}{10} = \tan\theta$ ... (1)



in Hz. direction $\mu N\cos\theta = N\sin\theta$
$\tan\theta = \mu = \frac{1}{2}$ ... (2)
from equation (1) and equation (2)
$\frac{x}{10} = \frac{1}{2}$ or $x = 5$m
$\Rightarrow y = \frac{x^2}{20} = \frac{25}{20} = \frac{5}{4} = 1.25$ m

35)

Question Explanation :
Determining the instantaneous acceleration of masses after cutting of the string.

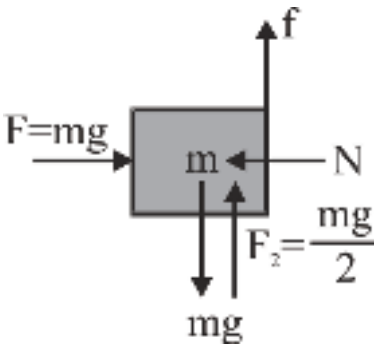
Concept: This question is based on cutting of string.

Solution:
Before the string is cut, the system is in equilibrium.
The spring forces (F_S) pulls the 2 m mass upwards, balancing total weight
F_S = (2 m + m) = 3 mg.
Just after the cutting of string, the tension in string becomes zero and lower block of mass fall freely under effect of gravity with acceleration g (downwards)
but the spring forces remains 3 mg
Net forces is F_S – 2 mg = 3 mg – 2 mg = 2ma (upward)
Acceleration of mass 2m $a_{2m} = \frac{mg}{2m} = \frac{g}{2}$ (upwards)
Final answer: Option (1)

36) **Explain Question:**
We have to calculate minimum value of coefficient of friction.

Concept:
Block against vertical surface.

Solution:



Here net driving force
$= mg - \frac{mg}{2} = \frac{mg}{2}$ downward
Hence friction will act upward and its magnitude should be $f = \frac{mg}{2}$. If the block 'm' is stationary the friction between m and the wall should be static.
$f \le f_{lim}$
$\frac{mg}{2} \leqslant \mu . N \Rightarrow \frac{mg}{2} \le \mu mg \Rightarrow \mu \ge \frac{1}{2}$
Final Answer: Option (3)

37) **Question Explanation**:
The lift is moving upward with speed 10 m/s, not acceleration. Since the speed is constant, the acceleration of the lift is zero.

Concept:
Apparent weight,

Solution:
Given

m = 50 kg

a= 0 (constant velocity)

Normal reaction: N = mg 50 × 10 500N

Reading in kg –wt: $\text{Reading} = \frac{N}{g} = \frac{500}{10} = 50\text{kg} - -\text{wt}$

Final answer: (1)

38)

$a = \frac{F_{net}}{m_{total}} = \frac{2\mu xg}{2L\mu} = \frac{xg}{L}$

39) $\vec{a}_A = 6t\hat{i} - 8t^5\hat{j}$

at t = 1 sec ; $\vec{a}_A = 6\hat{i} - 8\hat{j}$

$\therefore$ Pseudo force on B w.r.t. A} $= -m_B\vec{a}_A$

$= -50\left[6\hat{i} - 8\hat{j}\right] = 50\left[8\hat{j} - 6\hat{i}\right] = 100\left[4\hat{j} - 3\hat{i}\right]$

40) $a_1 = \frac{F}{m_1}\ a_2 = \frac{F}{m_2}$

$a = \frac{F}{m_1 + m_2} = \frac{F}{\frac{F}{a_1} + \frac{F}{a_2}} = \frac{a_1 a_2}{a_1 + a_2}$

$a = \frac{3 \times 6}{3 + 6} = 2\text{m/s}^2$

41)

Impulse = $\Delta p = m(v_2 - v_1)$

$= 70\left(\sqrt{2gh} - 0\right)$

$= 70\left(\sqrt{2 \times 9.8 \times 0.8} - 0\right)$

$= 70 \times 3.96$

$= 277.2$ kgm/sec

42) **Explain Question :** We need to determine the ratio of forces acting on particles A and B based on the slopes of their *p-t* (momentum-time) graph.

Concept : This question is based on force and momentum.

Formula : $F = \frac{dp}{dt} = \tan\theta$

Solution :

The ratio of forces acting on particles A and B is equal to the ratio of the slopes of their respective p-t graphs:

$\frac{F_A}{F_B} = \frac{\tan(\theta_A)}{\tan(\theta_B)}.$

For particle A: $\theta_A = 30^{\circ}$,

For particle B: $\theta_B = 45°$.

$\frac{F_A}{F_B} = \frac{\tan(30^{\circ})}{\tan(45^{\circ})} = \frac{\frac{1}{\sqrt{3}}}{1} = \frac{1}{\sqrt{3}}.$

$F_A : F_B = 1 : \sqrt{3}.$

Final Answer : Option (4)

43)

Explain the Question : For a body of **mass m = 2 kg** initially at rest, subjected to the **time-varying force (F−t) graph** shown, **find its final velocity v_2 after t = 15 sec**.

Concept : This question is based on the impulse - change in its linear momentum.

Formula : Impulse (J) = $\Delta p = m(v_2 - v_1)$ = Area of the F−t graph

Solution : Impulse = Δp = Area of force-time graph

$m(v_2 - v_1) = \frac{1}{2} \times 2 \times 10 + 3 \times 10 + \frac{1}{2} \times 5 \times 10 + 10 \times 5 + 2 \times 10 + \frac{1}{2} \times 3 \times 10$

$2(v - 0) = 10 + 30 + 25 + 50 + 20 + 15$

$v = 75$ m/sec

Final answer : 1

44)

Question Explanation : A single force **F** applied to a body of **mass M** produces an acceleration **a**. find the acceleration produced when **three forces, each equal to F and inclined to each other at 120°**, act on the same body (all the forces act in the same plane).

Concept : This question is based on Newton's Second Law.

Formula : $F_{net} = Ma'$

Solution :
Due to this **perfect symmetry**, the forces balance each other out completely.
The **net resultant force** is zero.
$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$
Newton's Second Law to the final state:
$F_{net} = Ma' = 0$
$a' = 0$
Final answer : option (4)

45) **Question Explanation:** For given system we need to calculate extension in spring.

Concept: Motion of connected bodied with pulley system with spring.

Solution:

For system

$$a_{system} = \frac{F_{net}}{M_{total}}$$

$$a_{system} = \frac{4g}{6+4} = 4m/s^2$$

Now for 6 kg block in Hz direction.

T = 6(a)

T = 6(4) = 24 N

so kx = 24

$$x = \frac{24}{k} = \frac{24}{240} = \frac{1}{10}m = 10 \text{ cm}$$

Final answer: option (2)

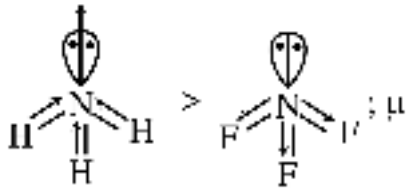
CHEMISTRY

46)

in SiO_2 π bond is absent between Si and O.

47)

NH_3 – N is sp^3 hybridised but in PH_3, no hybridisation

48) 

49) I_3^- ⇒ sp^3d hybridisation linear shape
N_3^- ⇒ sp hybridisation linear shape

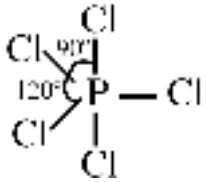
50)

	NO_2^+	NO_2^-	NO_3^-
N – O	2	1.5	1.3
B.O.			

$NO_2^+ < NO_2^- < NO_3^-$; N – O b.L.

51)

In PCl_5 Axial P – Cl bonds are longer than equitorial P – Cl bonds



52)

NH_3 NF_3 PH_3 PF_3
$NH_3 > NF_3 > PF_3 > PH_3$; order of bond angle

53) $\mu \propto \frac{1}{\theta}$

54)

Bond order of N_2^+ and N_2^- is 2.5.

55)

Between atoms of inert gas instantaneous – dipole induced dipole force is present.

56) H_2O > HF > NH_3 > CH_4 : boiling point
4H-bond 2H-bond 2H-bond VWF

57)

Fact based

58) B_2, O_2, O_2^+, O_2^- are paramagnetic

59)

Fact based

60)

In structure III □p-bp repulsion at 90° = 6 (maximum)

61)

Isostructural species

62) $\sqrt{v} \propto z$

63)

E has valence e^- config^n ⇒ $3s^2 3p^4$

64)

F^- < O^{-2} < N^{-3}; order of ionic radius

65)

B < Be < O < N; IE_1 order

66)

Be < Mg > Al > Si is correct order of radius

67)

A & B both are alkaline earth metal and B has large size than A.

68) $H_{(g)} + e^- \rightarrow H^-_{(g)}$ is exothermic in nature while remaining are endothermic in nature

69)

Order of Δng(–ve) for halogen ⇒ F < Cl > Br > I

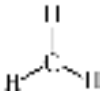
70)

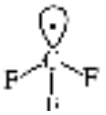
NO, N_2O , CO are neutral and remaining are acidic

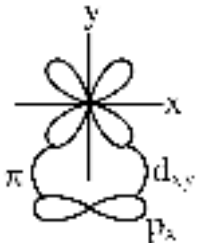
71)

$NH_3 < PH_3 < AsH_3 < SbH_3$; Acidic nature order

72)

CH_3 ⇒ sp^2, trigonal planer 

CF_3 ⇒ sp^3 pyramidal 

73) 

d_{xy} + p_x ⇒ π bond
INA will be y axis

74)

Explanation Question:
The question asks to determine if the equilibrium constant K_c is greater than, or equal to 1, based on the graph.
Concept:
This question is based on K_c the ratio of product concentrations to reactant concentration at equilibrium.
Solution:
The graph shows that at equilibrium, the concentration of reactant A is 0.1 M and the concentration of product B is 0.4 M. For the reaction A(g) ⇌ 2B(g),

$$K_c = \frac{[B]^2}{[A]}$$

1. Calculate K_c:

$$K_c = \frac{(0.4)^2}{0.1} = \frac{0.16}{0.1} = 1.6$$

Since 1.6 > 1, the equilibrium constant is greater than 1. This means product are favored, which is consistent with the graph showing a higher concentration of B than A at equilibrium.
Final Answer: (1)

75) **Explain the question :-** We need to find value of K_C for equilibrium reaction if degree of dissociation of PCl_5 is given.
Concept : Degree of dissociation and K_C.
Solution :

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

t=0 1 0 0
t_{eq} 1 – x x x

$\%\alpha = 80\%$; $80 = \frac{x}{1} \times 100 \Rightarrow = 0.8$

$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.8 \times 0.8}{1-0.8} = 3.2$

$K_C = 3.2$
Final answer : 4

76)

Reaction shifts backwards & reduced volume causes increase in concentration

77)

$2A \rightleftharpoons 2B + C$

1–α α $\frac{\alpha}{2}$

$K_P = \frac{(\alpha P)^2 (\frac{\alpha}{2} P)}{(P)^2}$

78)

$N_2 + 3H_2 \rightleftharpoons 2NH_3$
1 3
1–x 3–x 2x
$P_T = 3 = (1 - x) + (3 - x) + 2x$

79)

$CO + Cl_2 \rightleftharpoons COCl_2$
↓
rxn ← ∵ Cl_2 ↑ & $COCl_2$ ↓
$PCl_5 \rightleftharpoons PCl_3 + Cl_2$
↑
rxn ←
∵ PCl_5 ↑ & PCl_3 ↓

80)

$A(s) \rightleftharpoons B(g) + C(g)$
P_1 $P_1 + P_2$
$D(s) \rightleftharpoons C(g) + E(g)$

$P_1 + P_2 \quad P_2$

$x = P_1(P_1 + P_2)$; $y = (P_1 + P_2)(P_2)$

81)

Re x^n having $\Delta ng = 0$
are unaffected by pressure, volume and inert gas.

82) **Question Explanation :** Determine temperature and pressure conditions that favor formation of SO_3 in given reaction.
Concept : Le Chatelier's principle: exothermic reactions favor low T, fewer moles favor high P.
Solution:
$\Delta H <$ → reaction is exothermic → low temperature favors forward reaction.
Reactants: 2 + 1 = 3 moles; Products: 2 moles → high pressure favors forward reaction.
Final answer : option (1)

83) $\frac{K_p}{K_c} = (RT)^{\Delta n_g}$

$\log \frac{K_p}{K_c} - \Delta n_g \log RT = 0$

$\Delta n_g = -2$

84)

$K_P = K_C (RT)^1$
$3 = RT$
$T = \frac{3}{R} = 3 \times 12.18$

85)

$A(s) \rightleftharpoons 2B(g) + C(g)$
$\quad 2x \quad x + 0.5$
$0.5 + 2x + x = 0.8$
$x = 0.1$
$K_P = P_B^2 \times P_C$
$= (0.2)^2(0.6)$
$= 0.024$

86)

Mixture of CCl_4 & benzene shows ⊕ ve deviation

87)

P_S (theoretical)

$= P_A^o X_A + P_B^o X_B$

$= 150 \times \frac{1}{3} + 240 \times \frac{2}{3}$

$= 50 + 160 = 210$

$P_{S\ Theoretical.} > P_{S\ Obs}$

 negative deviation

88) By Henry's law,

For dilute solutions

$p_{O_2} = k_H X_{O_2}$

$0.9 = 1.6 \times 10^4 \times \frac{n_{O_2}}{n_{H_2O}}$

$0.9 = 1.6 \times 10^4 \times \frac{w_{O_2}}{32} \times \frac{18}{100}$

$w_{O_2} = 0.01g$

89)

As T ↑ ⇒ K_w ↑ ⇒ pK_w ↓

 pH & pOH ↓

90)

Solution:

Initial $[H^+] = 10^{-4}$ mol/dm^3

After mixing equal volume of water, new $[H^+] = 10^{-4} / 2 = 0.5 \times 10^{-4}$ mol/dm^3

$[OH^-] = K_w / [H^+] = 10^{-14} / (0.5 \times 10^{-4}) = 2 \times 10^{-10}$ mol/dm^3

Final Answer: (2)

BIOLOGY

91)

Midgut and foregut

92) 3

93)

NCERT Pg # 111

94)

NCERT Pg # 112

95) Module Pg.# 161,162

96)

NCERT Pg # 115

97)

NCERT Pg # 42

98)

NCERT Pg # 46

99) NCERT Pg. # 48

100) NCERT Pg. # 49

101) **Question Explanation:** Incorrect feature for Pleurobrachia
Concept: Comb plate count
Solution: The comb jelly Pleurobrachia has eight comb plates, not ten. The other options are correct: it shows bioluminescence, reproduces only sexually, and has tissue-level organization.
Final Answer: Presence of ten comb plate

102)

NCERT Pg # 42

103) NCERT Pg #44

104)

NCERT Pg # 38

105)

NCERT Pg # 49

106)

NCERT Pg # 45

107)

NCERT Pg # 44

108)

NCERT Pg # 41

109) NCERT–XI, Pg. # 56

110)

NCERT Pg # 49

111) NCERT, Pg. # 37 to 52

112) NCERT-XI, Pg. # 41,43,46,47

113)

NCERT Pg # 42

114)

Explanation :
Adamsia (1) - Sea anemone (b) - Polyp only (iv)

Aurelia (2) - Jellyfish (d) - Medusa only (i)

Obelia (3) - (a) - Polyp and medusa both (ii)

Pleurobrachia (4) - Sea gooseberry (c) - no polyp and medusa (iii)

Final Answer : (4)

115) NCERT, Pg. # 41

116)

NCERT Pg # 48

117) NCERT, Pg. # 45

118)

NCERT Pg # 42, 43

119) NCERT-XI, Pg.# 56,57

120) NCERT, Pg. # 48, 49, 50

121) NCERT, Pg. # 42

122) NCERT-XI, Pg. # 49-51

123)

NCERT Pg # 118

124)

NCERT XI, Pg. No. 116

125)

NCERT PAGE NO. 81, 82

126)

NCERT Pg # 116

127) NCERT Pg # 80-81

128) NCERT XI, Pg. # 116, Para - 7.5.1

129) NCERT Pg # 80

130) NCERT Pg # 81

131) NCERT Pg # 82

132) NCERT Pg # 82

133)

NCERT Pg#117

134) NCERT Pg. # 81, 82

135) NCERT-XI Pg. # 116

136) NCERT-XI Pg. # 118

137) NCERT-XI Pg. # 40

138)

NCERT Pg # 43, 45, 50

139) NCERT Pg. # 104

140)

NCERT Pg # 103

141) NCERT, Pg. # 102

142)

NCERT Pg # 105

143) **Solution/Explanation/Calculation:**
Explanation:
Here's the correct matching:

A. (A) Unicellular Gland → (i) Goblet cell
Goblet cells are unicellular glands that secrete mucus.
(B) Tight Junction → (iv) Help to stop substances from leaking across tissue
Tight junctions seal adjacent epithelial cells to prevent leakage.
(C) Gap Junction → (ii) For rapid transfer of ions and small molecules
Gap junctions allow communication between cells, enabling the passage of ions, nutrients, and signaling molecules.
(D) Adhering junction → (iii) Perform cementing to keep neighbouring cells together
Adhering junctions provide mechanical stability by anchoring cells together.

Correct Answer:
Option (1) – A-i, B-iv, C-ii, D-iii

144)

NCERT Pg # 100

145) NCERT-XI Page. 103

146) NCERT-XI, Pg. # 112

147) NCERT, Pg. # 109
Concept :
Cell composition → Tissue analysis
Solution :
Lipids → 2%, Ions → 1%
Carbohydrates → 3%
Nucleic Acid → 5-7%
Final Answer :
Option (2)

148) NCERT-XI Pg. No. # 105,108,110,111

149) NCERT XI Page No. # 112

150)

NCERT # XI, Page No. 117

151) NCERT Pg. # 112, 113

152) NCERT (XIth) Pg. # 113

153) NCERT Pg.# 116

154) NCERT-XI, Page No. # 135

155) NCERT Pg. # 140

156) NCERT, Pg. # 144

157) NCERT, Pg. # 146, 143

158)

NCERT XI Pg. # 150

159) NCERT Pg # 147

160) NCERT XI, Pg.No. # 136

161) NCERT, Pg. # 145
bundle sheath cells don’t have large intercellular spaces

162) NCERT (XI) ; Page No. # 133, 136

163)

NCERT, Pg. # 156

164)

NCERT XI Pg.# 158, 159

165) NCERT Pg. # 160

166)

NCERT XI, Pg. # 162, 163

167)

NCERT-XI, Pg. # 158
Question Explanation
The question is asking whether the assertion and reason about the TCA cycle are true, and whether the reason correctly explains the assertion.

Concept
TCA cycle – reactions and location.

Solution
The assertion is true: the TCA cycle begins with condensation of acetyl-CoA with oxaloacetate (OAA) and water to form citric acid. The reason is also true: the TCA cycle occurs in the mitochondria. However, the reason does not explain why the cycle starts with condensation; it only states the location.

Final answer: Option 2

168) NCERT-XI, Pg. # 159

169) NCERT-XI, Pg # 154,155,156, 159

170) NCERT-XI, Pg. # 155-156

171) NCERT-XI Pg. # 154, 157, 158, 160

172)

NCERT-XI Pg. # 177

173)

NCERT XI Pg # 177,178

174) NACET XI Pg. # 175

175) NCERT, Pg # 170

176) NCERT-XI, Pg. # 173-175

177) NCERT (XIth) Pg. # 171

178)

NCERT-XI, Pg. # 174
Carotenoids are accessory photosynthetic pigments.

179) NCERT-XI Page No.# 175-177

180)

NCERT XI, Pg # 177